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(54) **SEMICONDUCTOR DEVICE AND METHOD OF MANUFACTURING SEMICONDUCTOR DEVICE**

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(Continued)

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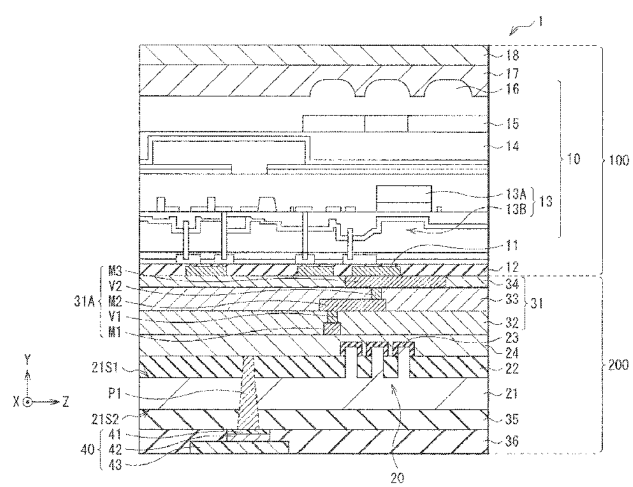
(57) **ABSTRACT**

A semiconductor device according to an embodiment of the present disclosure includes: a first substrate provided with an active element; and a second substrate laminated with the first substrate and electrically coupled to the first substrate, in which the second substrate is provided with a first transistor configuring a logic circuit on a first surface and with a non-volatile memory element on a second surface opposite to the first surface.

20 Claims, 15 Drawing Sheets



(51) **Int. Cl.**
H10F 39/00 (2025.01)
H10N 50/01 (2023.01)
H10N 50/10 (2023.01)

(52) **U.S. Cl.**
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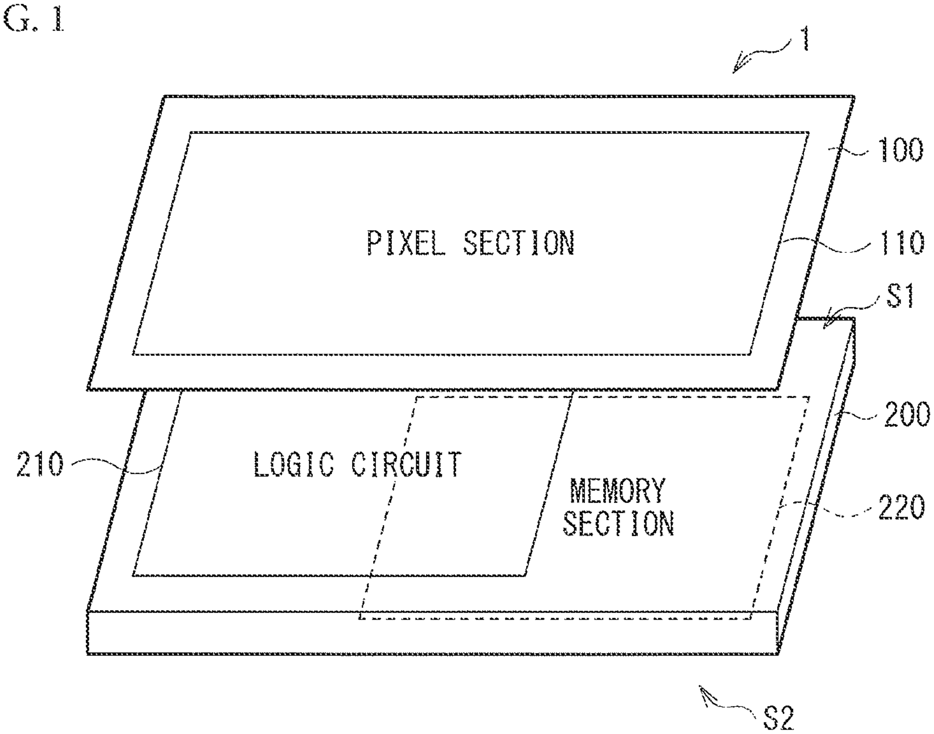
FIG. 1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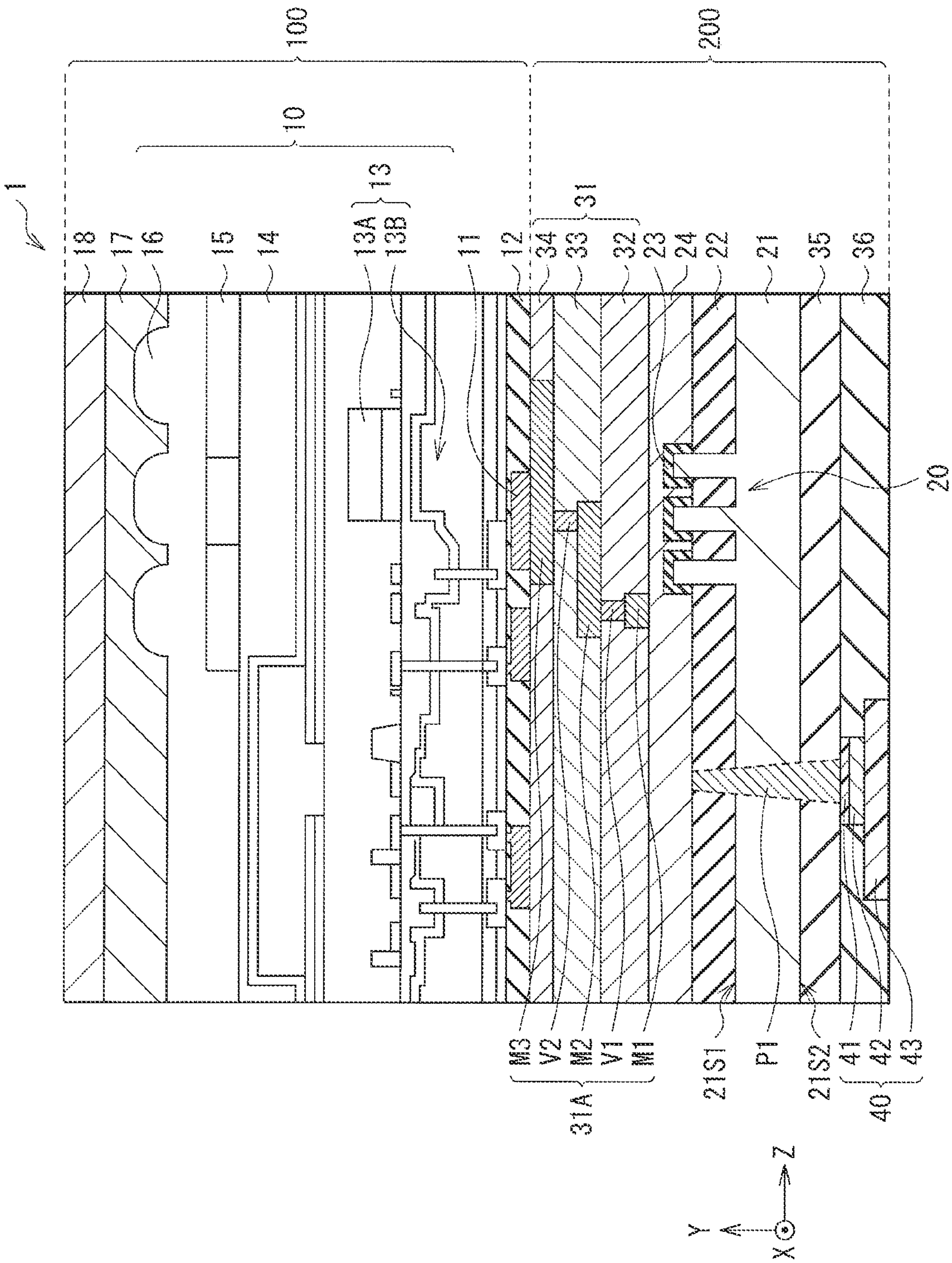
PIXEL SECTION
110
S1
200
210
LOGIC CIRCUIT
MEMORY
SECTION
220
S2

FIG. 2



100
200
10
1
18
17
16
15
14
13A
13B
13
11
12
34
33
32
31
23
24
22
21
35
36
20
M3
V2
M2
V1
M1
31A
21S1
P1
21S2
41
42
43
40
Z
Y
X

FIG. 3

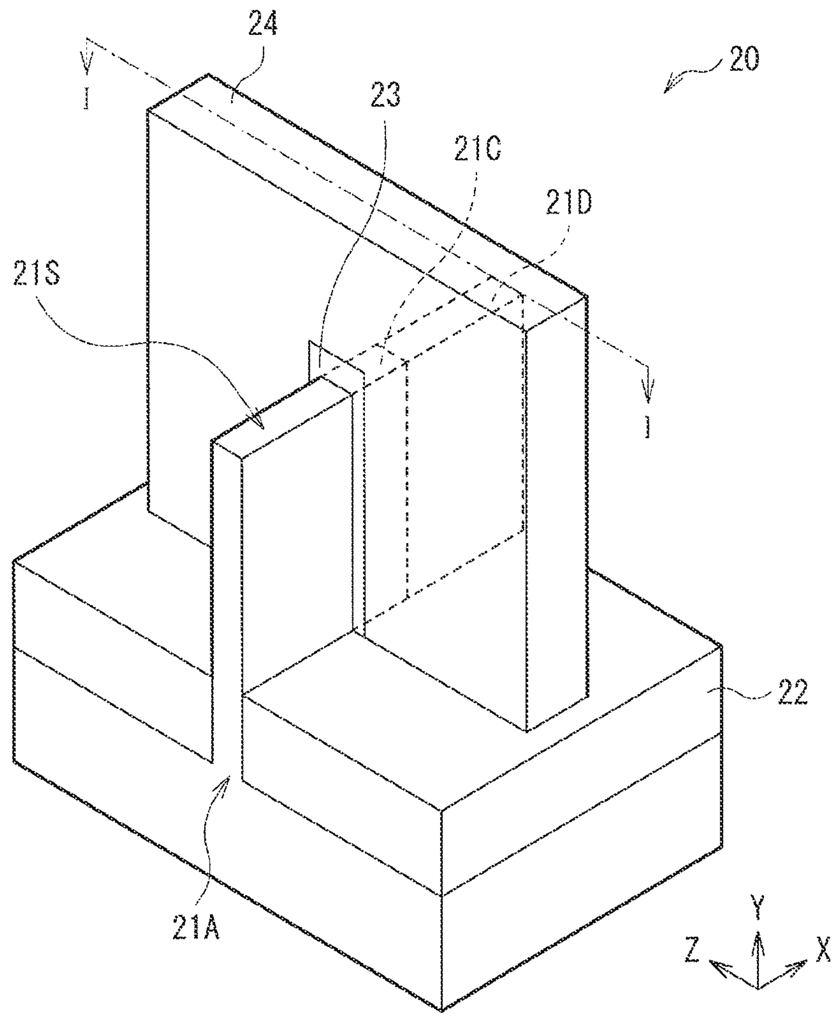

24
I
23
20
21C
21D
21S
I
22
21A
Y
Z
X

FIG. 4

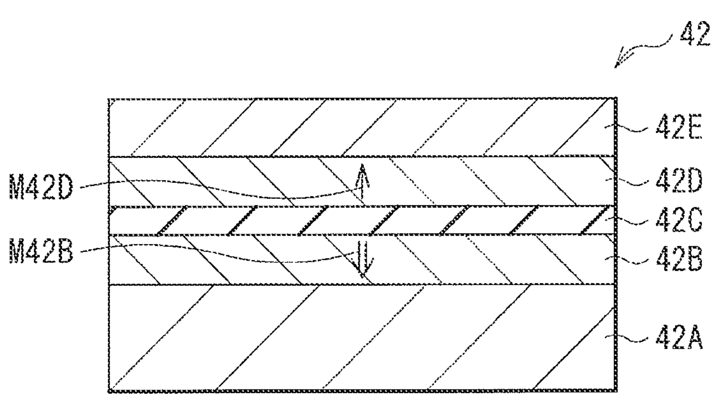
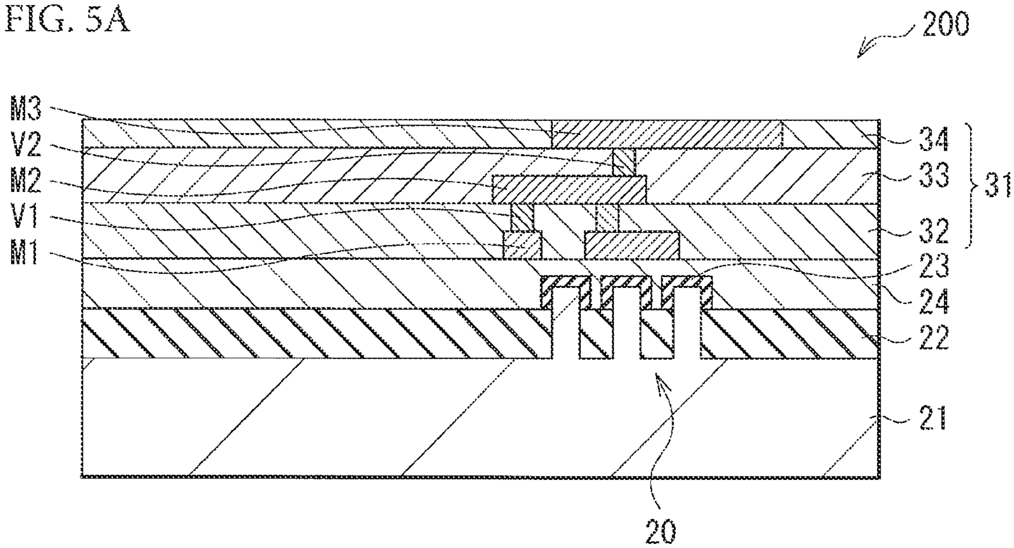

42
42E
M42D
42D
42C
M42B
42B
42A

FIG. 5A


200
M3
V2
M2
V1
M1
34
33
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23
24
22
21
20

FIG. 5B

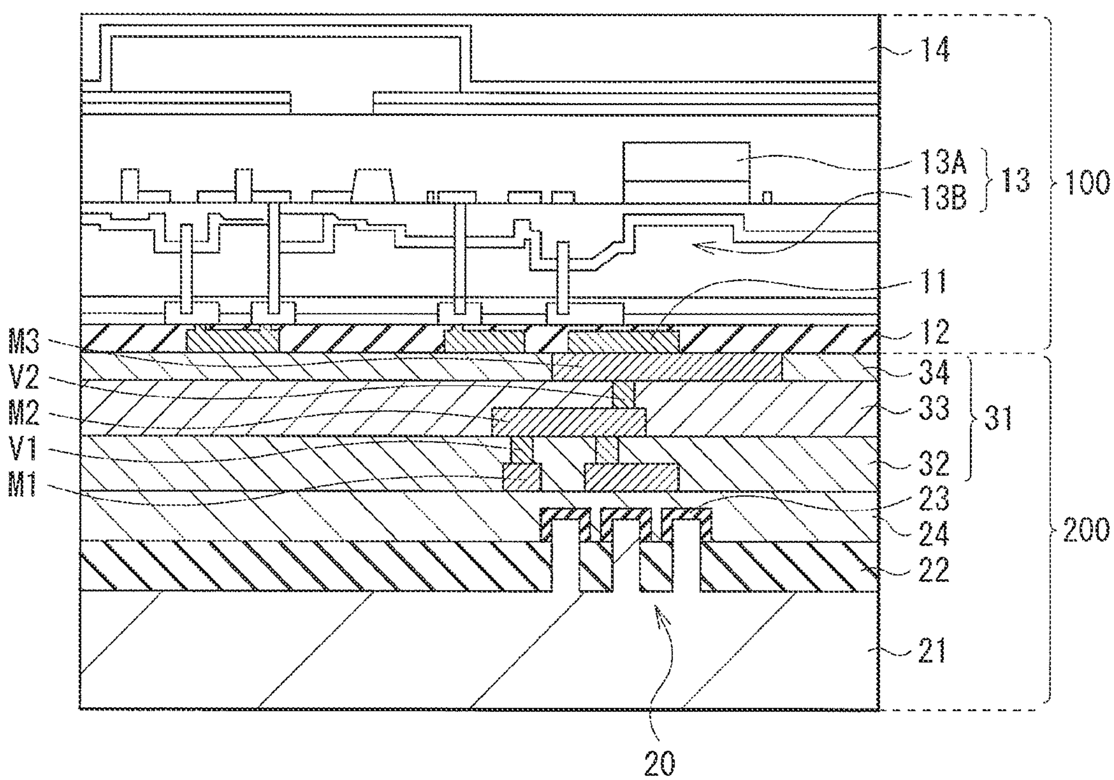
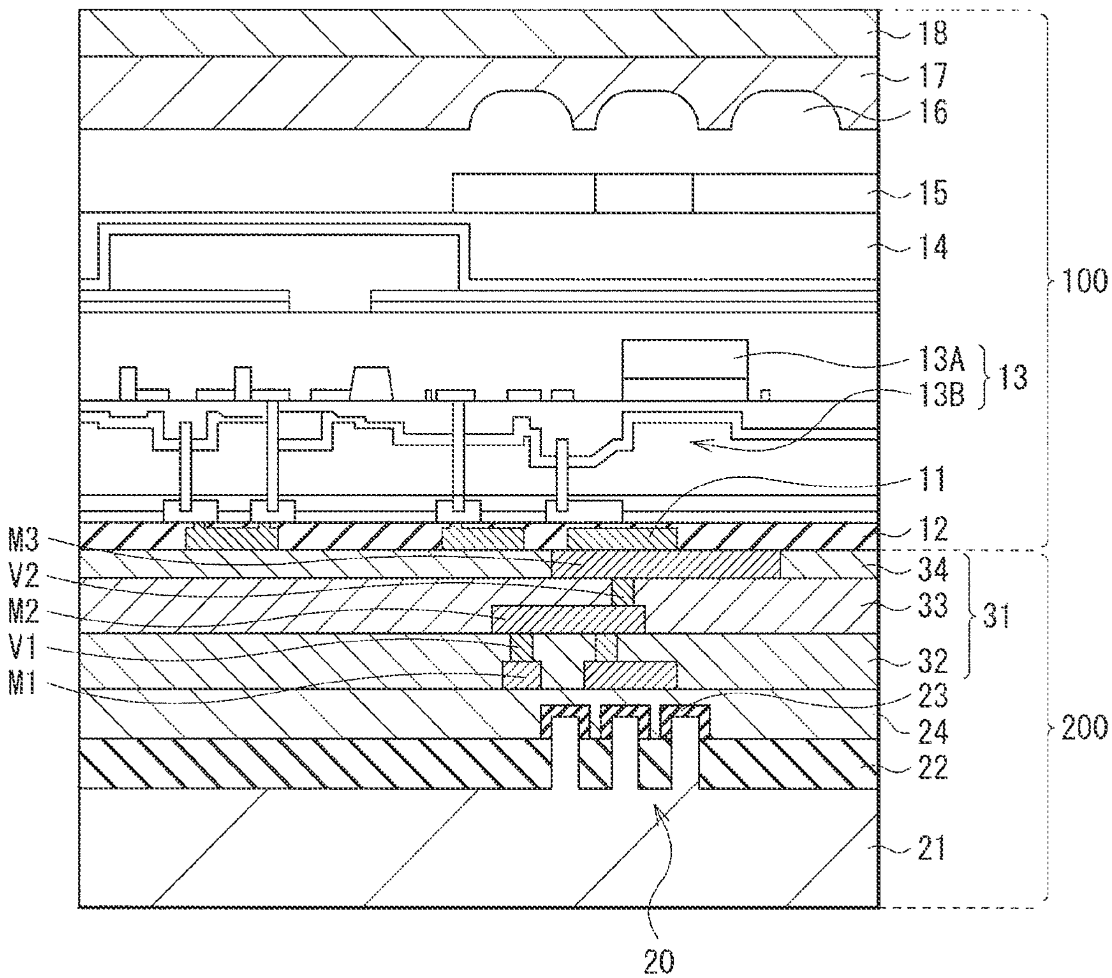

14
13A
13B
13
100
11
12
M3
V2
M2
V1
M1
34
33
31
32
23
24
22
200
21
20

FIG. 5C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17
16
15
14
100
13A
13B
13
11
12
M3
V2
M2
V1
M1
34
33
31
32
23
24
200
22
21
20

FIG. 5D

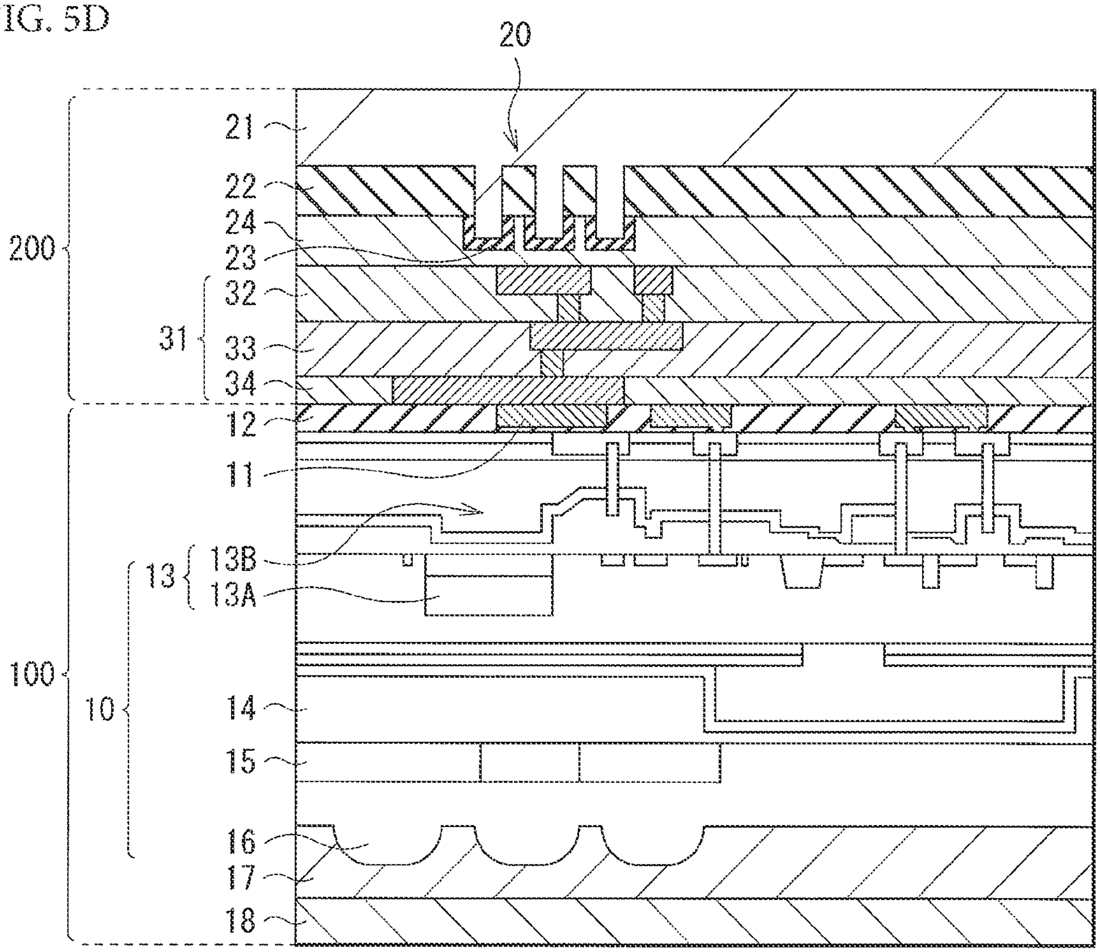
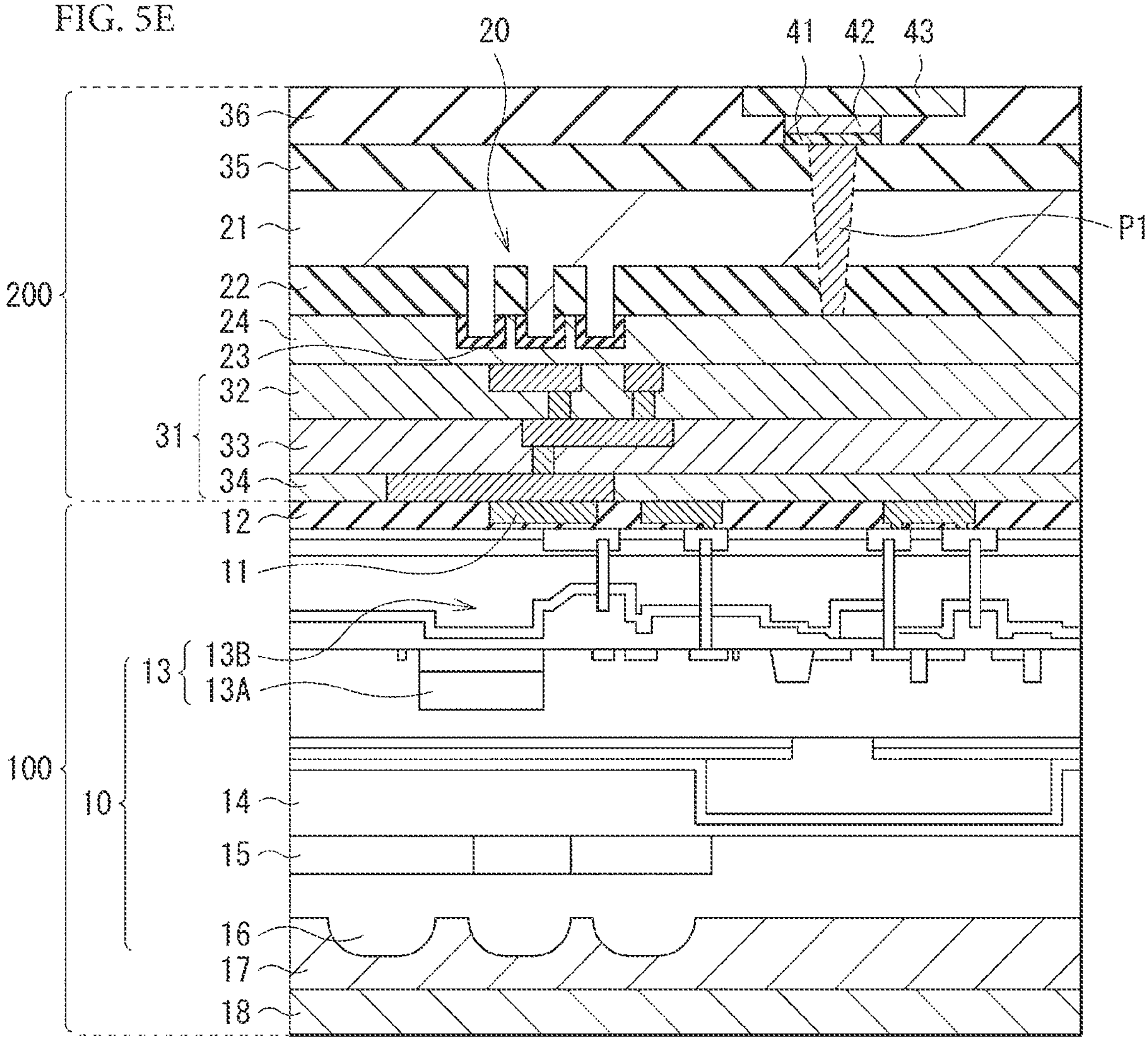


FIG. 5E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P1
22
24
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32
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34
200
12
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13B
13
13A
100
10
14
15
16
17
18

FIG. 6

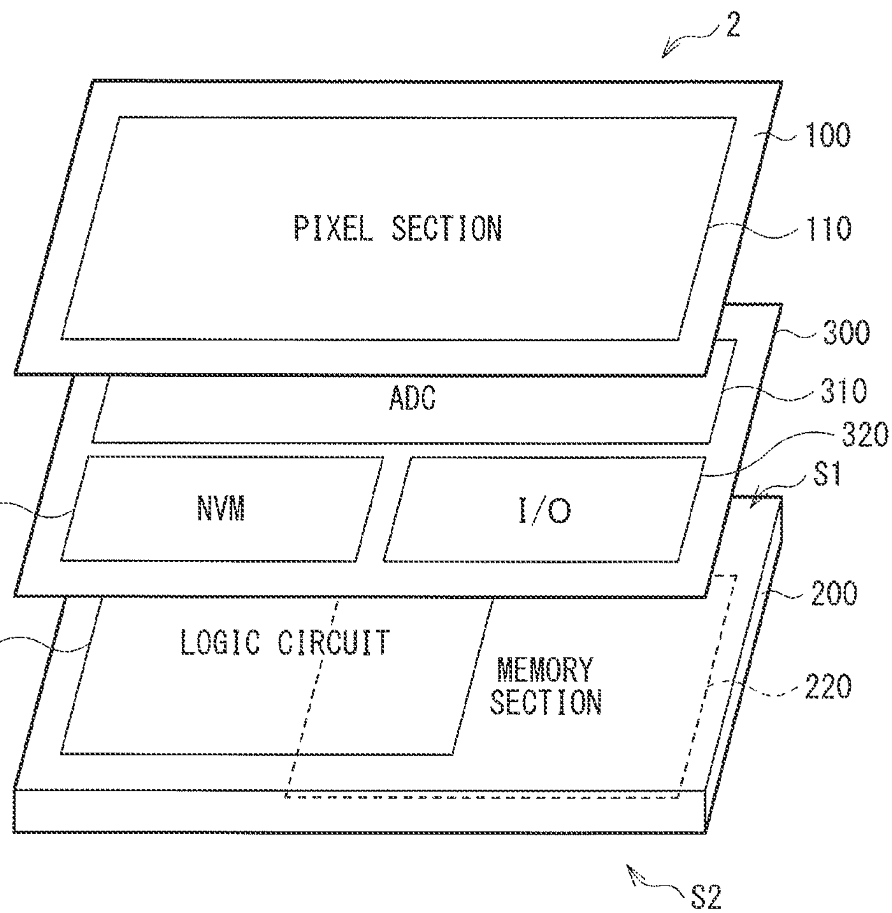

2
100
PIXEL SECTION
110
300
ADC
310
320
S1
330
NVM
I/O
200
210
LOGIC CIRCUIT
MEMORY
SECTION
220
S2

FIG. 7

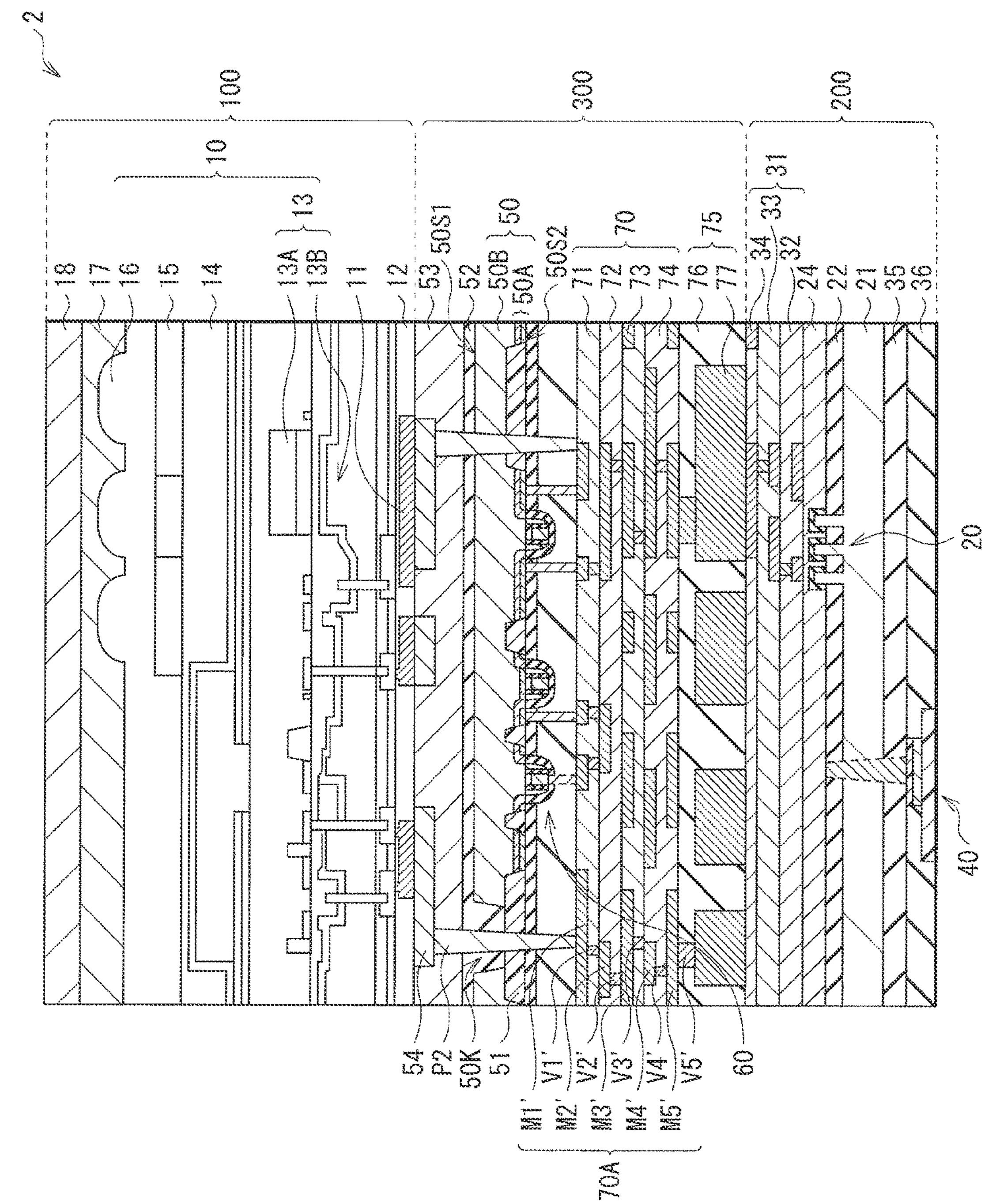
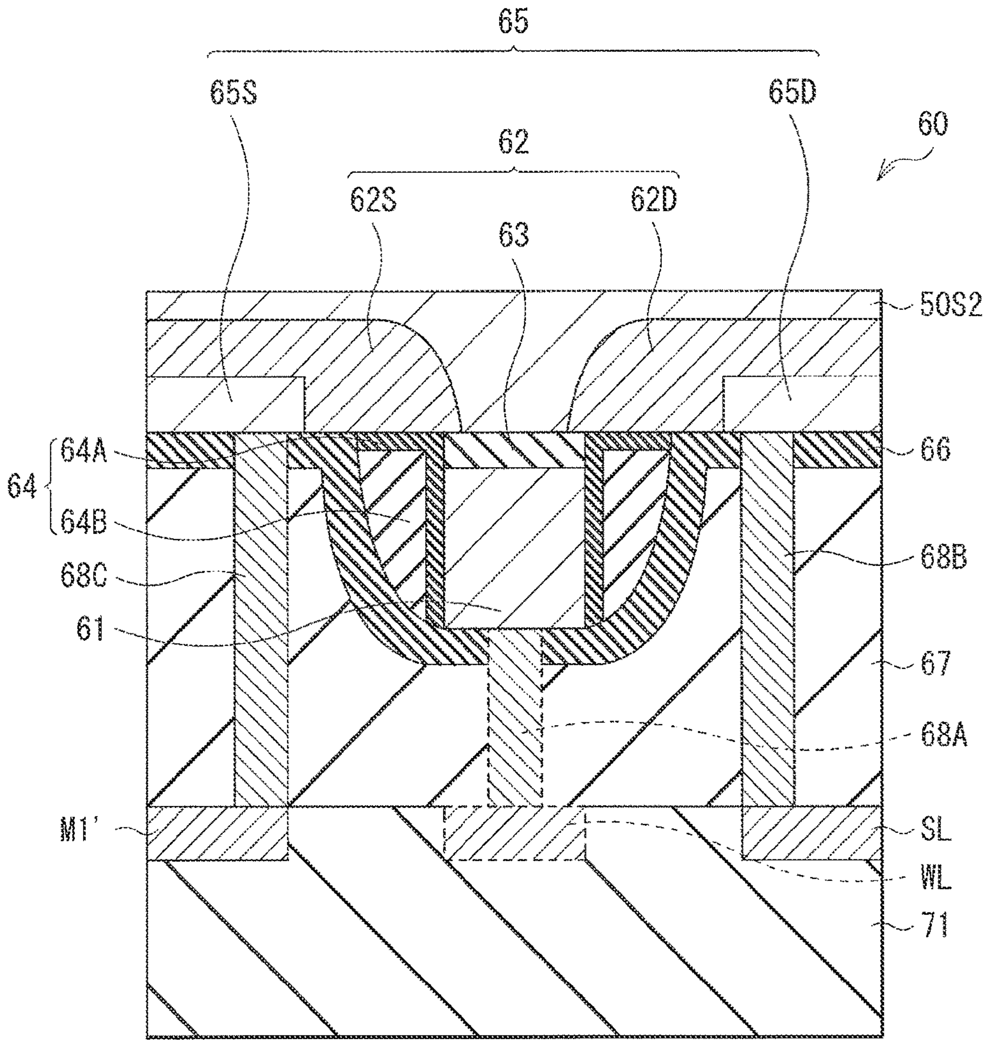

2
100
10
13
13A
13B
300
200
31
33
50
70
75
18
17
16
15
14
11
12
53
50S1
52
50B
50A
50S2
71
72
73
74
76
77
34
32
24
22
21
35
36
20
40
54
P2
50K
51
V1'
V2'
V3'
V4'
V5'
60
M1'
M2'
M3'
M4'
M5'
70A

FIG. 8


65
65S
65D
60
62
62S
62D
63
50S2
64A
66
64
64B
68C
68B
61
67
68A
M1'
SL
WL
71

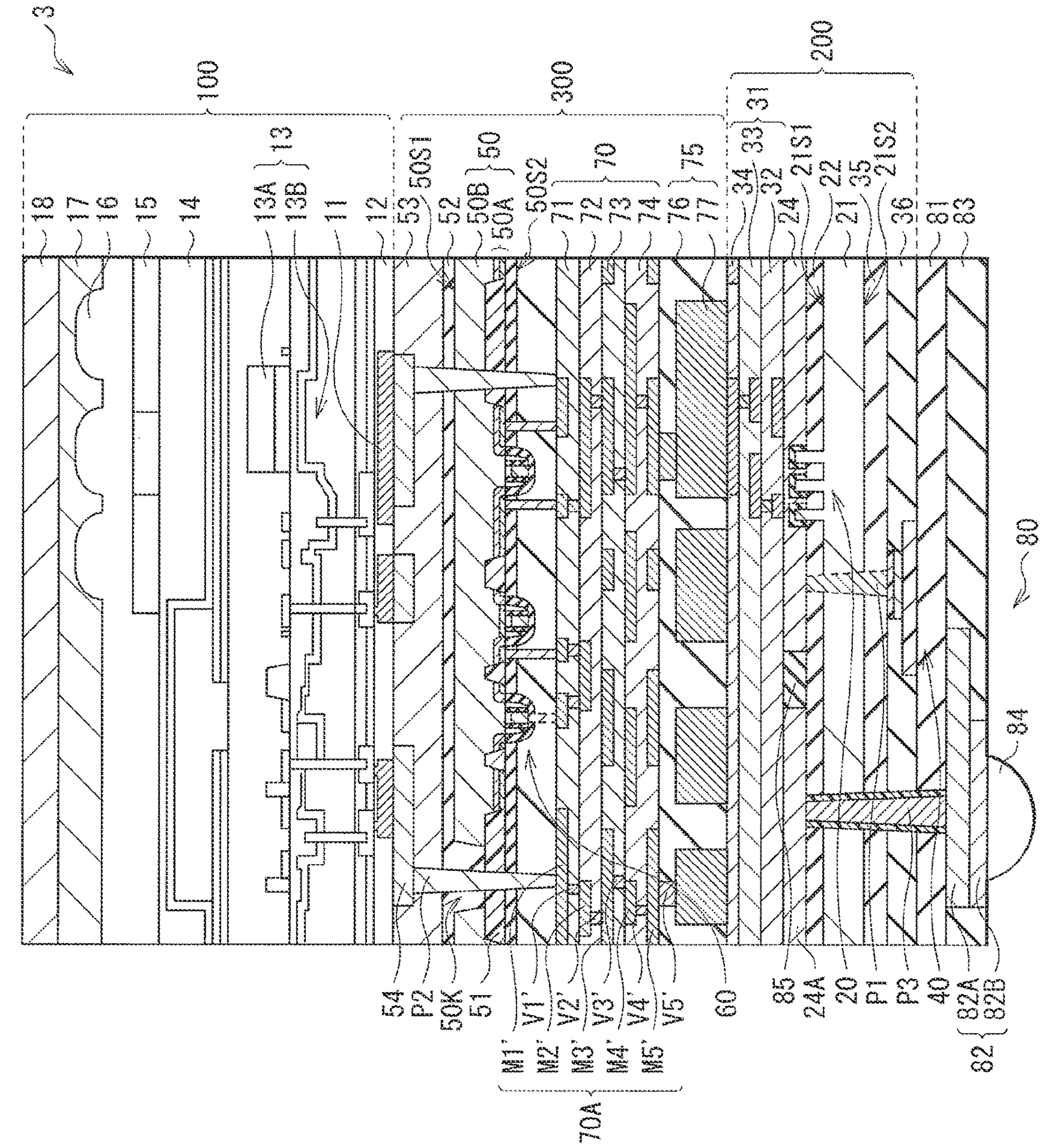

3
100
300
200
18
17
16
15
14
13A
13B
13
11
12
53
50S1
52
50B
50A
50
50S2
71
72
73
74
70
76
77
75
34
33
32
31
24
21S1
22
21
35
21S2
36
81
83
80
84
54
P2
50K
51
M1'
V1'
M2'
V2'
M3'
V3'
M4'
V4'
M5'
V5'
70A
60
85
24A
20
P1
P3
40
82A
82B
82

FIG. 9

FIG. 10

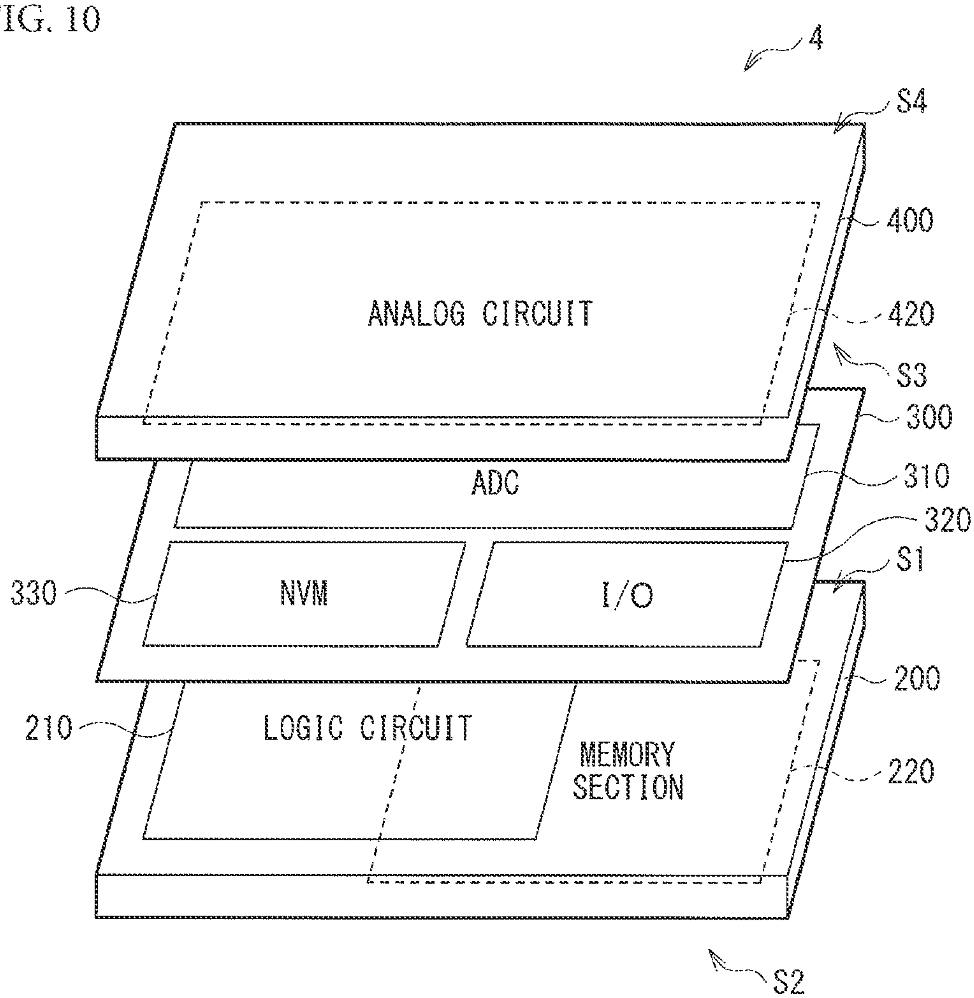
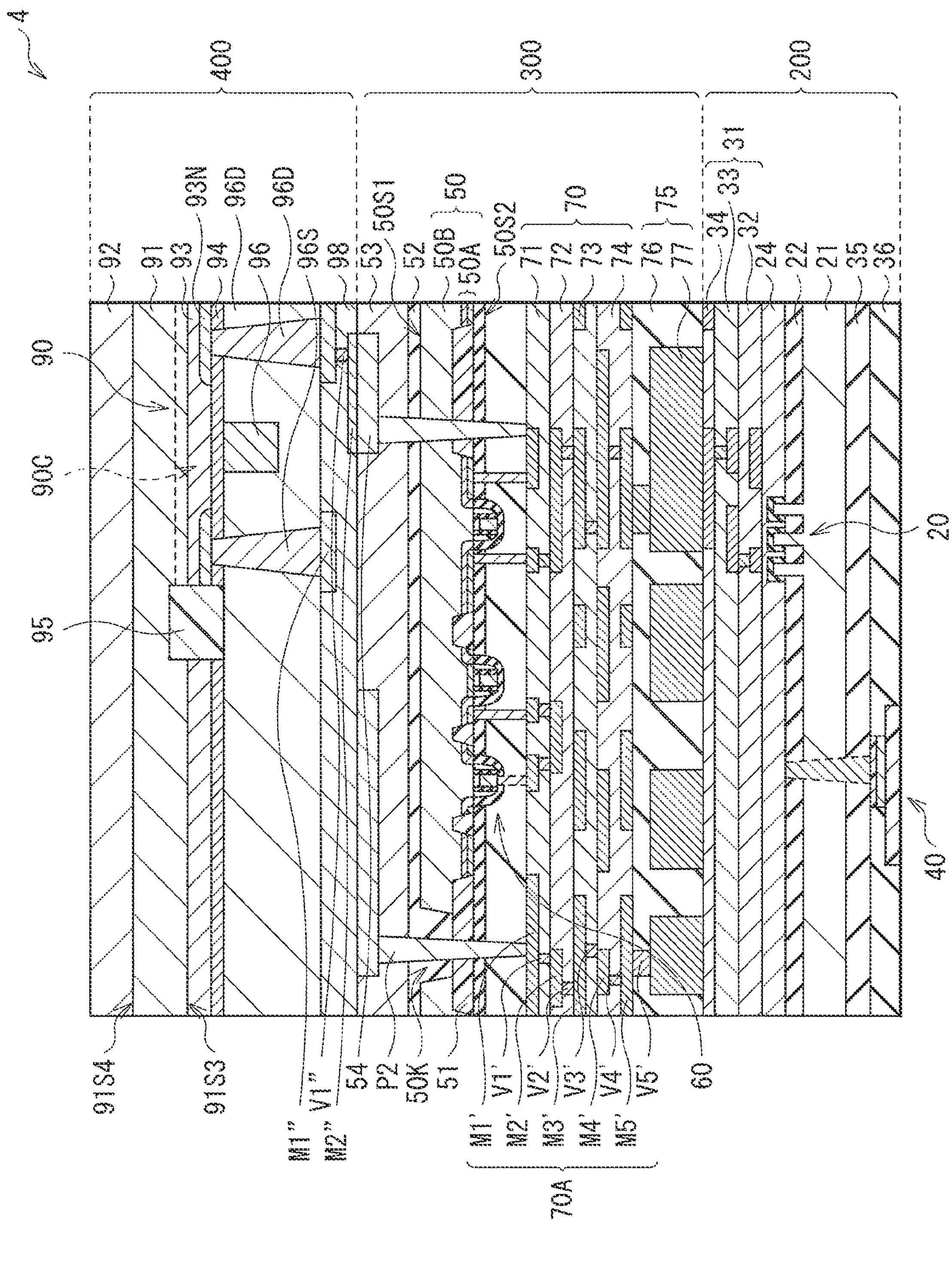

4
S4
400
ANALOG CIRCUIT
420
S3
300
ADC
310
320
S1
330
NVM
I/O
200
210
LOGIC CIRCUIT
MEMORY SECTION
220
S2

FIG. 11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400
300
200
92
91
93
93N
94
96D
96
96D
96S
98
53
50S1
52
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50A
50S2
71
72
70
73
74
76
75
77
34
33
31
32
24
22
21
35
36
90
90C
95
20
40
91S4
91S3
M1"
V1"
M2"
54
P2
50K
51
M1'
V1'
M2'
V2'
M3'
V3'
M4'
V4'
M5'
V5'
60
70A

FIG. 12

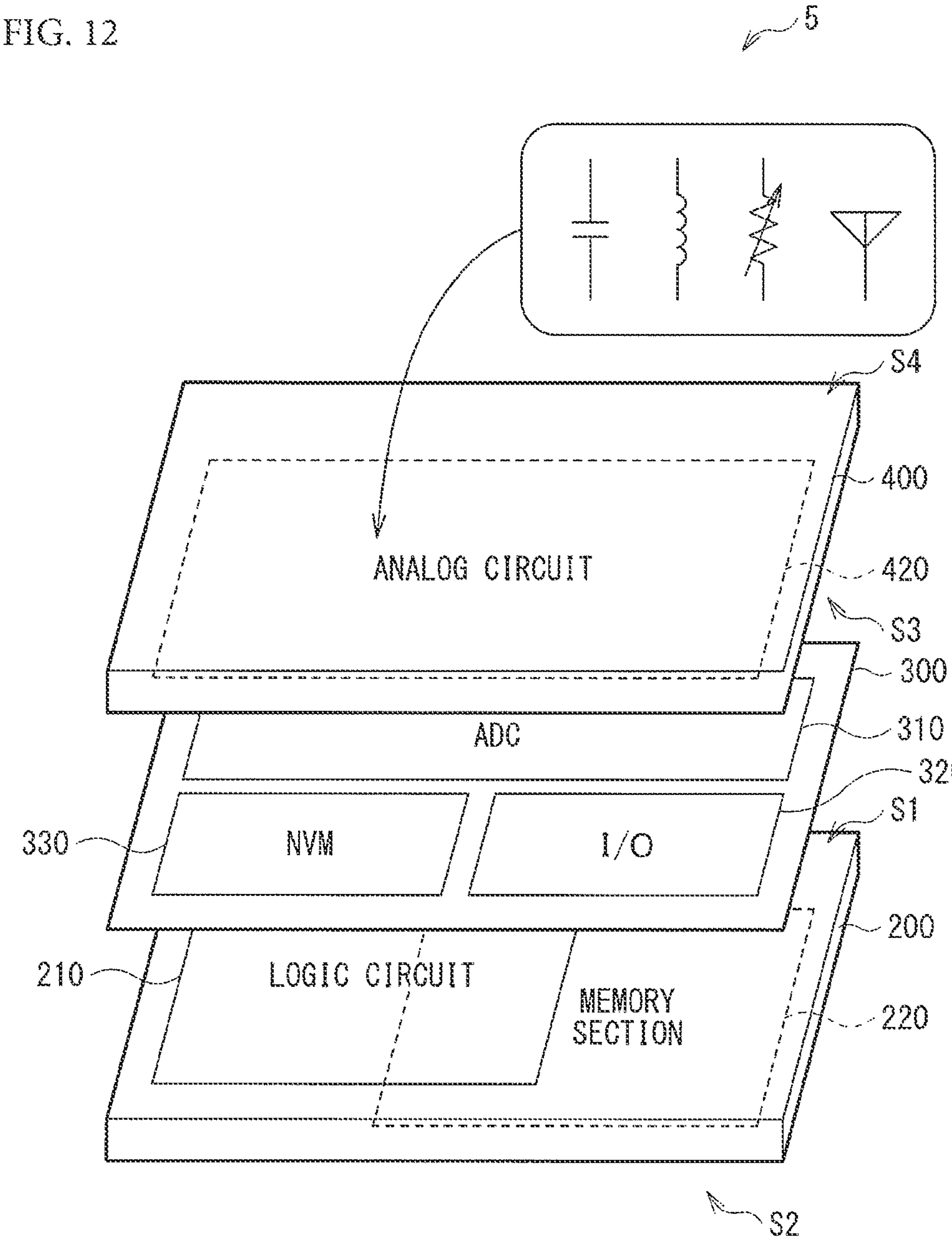
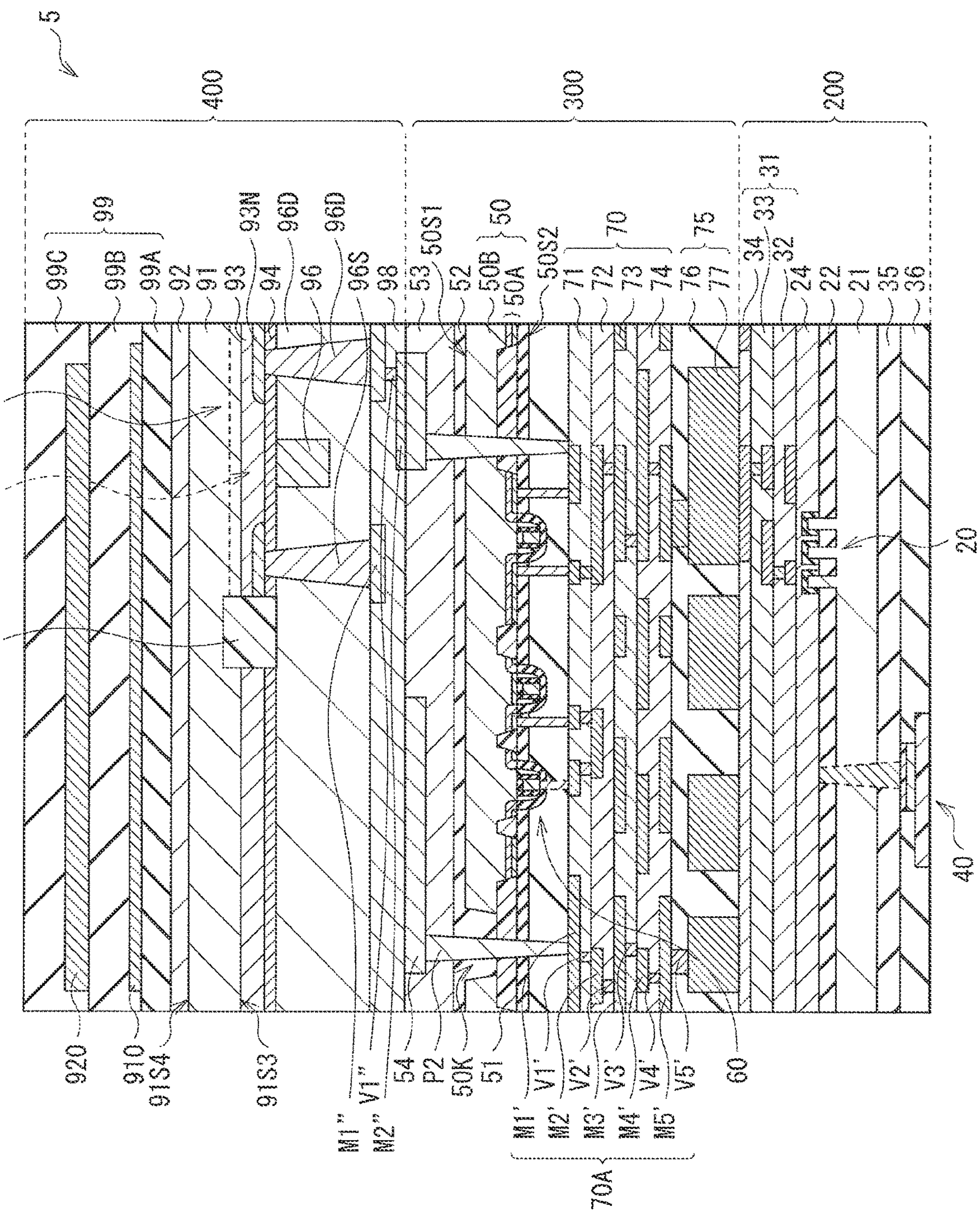

5
S4
400
ANALOG CIRCUIT
420
S3
300
ADC
310
320
S1
330
NVM
I/O
200
210
LOGIC CIRCUIT
MEMORY
SECTION
220
S2

FIG. 13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400
300
200
99
99C
99B
99A
92
91
93
93N
94
96D
96D
96
96S
98
53
50S1
52
50
50B
50A
50S2
70
71
72
73
74
75
76
77
34
33
32
31
24
22
21
35
36
90
90C
95
20
40
920
910
91S4
91S3
M1"
M2"
V1"
54
P2
50K
51
M1'
M2'
M3'
M4'
M5'
V1'
V2'
V3'
V4'
V5'
60
70A

SEMICONDUCTOR DEVICE AND METHOD OF MANUFACTURING SEMICONDUCTOR DEVICE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a national stage application under 35 U.S.C. 371 and claims the benefit of PCT Application No. PCT/JP2018/000408 having an international filing date of 11 Jan. 2018, which designated the United States, which PCT application claimed the benefit of Japanese Patent Application No. 2017-020626 filed 7 Feb. 2017, the entire disclosures of each of which are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a semiconductor device including, for example, a non-volatile memory element using a magnetic material, and a method of manufacturing the semiconductor device.

BACKGROUND ART

An MTJ (Magnetic Tunnel Junction) element that is a non-volatile memory element using a magnetic material has low resistance to heat. This may possibly lead to degradation due to thermal budget during a wiring forming process. To solve this, for example, PTL 1 discloses a method of manufacturing a semiconductor device by forming the MTJ element on side of a back surface of a substrate after completion of the wiring forming process.

Incidentally, PTL 2 discloses a semiconductor device including an image sensor laminated on a logic circuit.

CITATION LIST

Patent Literature

PTL 1: Japanese Unexamined Patent Application Publication No. 2014-220376

PTL 2: Japanese Unexamined Patent Application Publication No. 2015-65407

SUMMARY OF THE INVENTION

Thus, a semiconductor device that includes an image sensor laminated on a substrate including an MTJ element may possibly lead to degradation of characteristics of the MTJ element due to thermal budget during a lamination process of the image sensor. Therefore, it is desired to develop a method of manufacturing a semiconductor device that allows for preventing the degradation of the characteristics of a non-volatile memory element such as the MTJ element.

It is desirable to provide a semiconductor device that allows for preventing degradation of characteristics of a non-volatile memory element and a method of manufacturing the semiconductor device.

A semiconductor device according to an embodiment of the present disclosure includes: a first substrate provided with an active element; and a second substrate laminated with the first substrate and electrically coupled to the first substrate, in which the second substrate is provided with a first transistor configuring a logic circuit on a first surface and with a non-volatile memory element on a second surface opposite to the first surface.

A method of manufacturing a semiconductor device according to an embodiment of the present disclosure includes: forming an active element on a first substrate; forming a transistor configuring a logic circuit on a first surface of a second substrate; electrically coupling the first substrate and the second substrate together: and forming a non-volatile memory element on a second surface of the second substrate opposite to the first surface.

In the semiconductor device according to an embodiment of the present disclosure and the method of manufacturing the semiconductor device according to an embodiment of the present disclosure, the first transistor configuring the logic circuit is provided on the first surface of the second substrate, and the non-volatile memory element is provided on the second surface opposite to the first surface. This makes it possible to provide the non-volatile memory element at a desired timing and thus to reduce thermal budget applied to the non-volatile memory element.

Accordance to the semiconductor device according to an embodiment of the present disclosure and the method of manufacturing the semiconductor device according to an embodiment of the present disclosure, the first transistor configuring the logic circuit is provided on the first surface and the non-volatile memory element is provided on the second surface opposite to the first surface, thus making it possible to provide the non-volatile memory element at a desired timing. This makes it possible to reduce the thermal budget applied to the non-volatile memory element and thus to prevent degradation of characteristics of the non-volatile memory element.

It is to be noted that effects of the present disclosure is not limited to those described above, and may be any of the effects described below.

BRIEF DESCRIPTION OF DRAWING

FIG. **1** is a schematic view of a semiconductor device according to a first embodiment of the present disclosure.

FIG. **2** is a cross sectional view of a specific configuration of the semiconductor device illustrated in FIG. **1**.

FIG. **3** is an explanatory perspective view of a transistor provided on a second substrate of the semiconductor device illustrated in FIG. **2**.

FIG. **4** is a cross sectional view of a configuration of a storage section of a storage element provided in the second substrate of the semiconductor device illustrated in FIG. **2**.

FIG. **5**A is a cross sectional view that describes a method of manufacturing the semiconductor device illustrated in FIG. **2**.

FIG. **5**B is a cross sectional view of a process step that follows FIG. **5**A.

FIG. **5**C is a cross sectional view of a process step that follows FIG. **5**B.

FIG. **5**D is a cross sectional view of a process step that follows FIG. **5**C.

FIG. **5**E is a cross sectional view of a process step that follows FIG. **5**D.

FIG. **6** is a schematic view of a semiconductor device according to a second embodiment of the present disclosure.

FIG. **7** is a cross sectional view of a specific configuration of the semiconductor device illustrated in FIG. **6**.

FIG. **8** is a cross sectional view that describes a transistor included in a third substrate of the semiconductor device illustrated in FIG. **7**.

FIG. **9** is a cross sectional view of a semiconductor device according to Modification Example 1 of the present disclosure.

FIG. **10** is a schematic view of a semiconductor device according to a third embodiment of the present disclosure.

FIG. **11** is a cross sectional view of a specific configuration of the semiconductor device illustrated in FIG. **10**.

FIG. **12** is a schematic view of a semiconductor device according to Modification Example 2 of the present disclosure.

FIG. **13** is a cross sectional view of a specific configuration of the semiconductor device illustrated in FIG. **12**.

MODES FOR CARRYING OUT THE INVENTION

In the following, embodiments of the present disclosure are described in detail with reference to the drawings. The following description is directed to a specific example of the present disclosure, and the present disclosure is not limited to the following embodiments. Moreover, the present disclosure is not limited to positions, dimensions, dimension ratios, and other factors of components illustrated in the drawings. It is to be noted that the description is given in the following order.

1. First Embodiment
(An example semiconductor device including a first substrate and a second substrate laminated together, the first substrate having an image sensor and the second substrate including a logic circuit on a surface S**1** and a non-volatile memory element on a surface S**2**)
    1-1. Configuration of Semiconductor Device
    1-2. Method of Manufacturing Semiconductor Device
    1-3. Workings and Effects
2. Second Embodiment (An example semiconductor device including three substrates laminated together)
3. Modification Example 1 (An example in which a leading electrode is further provided on the surface S**2** of the second substrate)
4. Third Embodiment (An example semiconductor device provided with a circuit having a communication function on the first substrate)
5. Modification Example 2 (An example in which an antenna is further provided in addition to the circuit having the communication function)

First Embodiment

(1-1 Configuration of Semiconductor Device)

FIG. **1** illustrates a schematic configuration of a semiconductor device (semiconductor device **1**) according to a first embodiment of the present disclosure. The semiconductor device **1** includes a first substrate **100** and a second substrate **200** electrically coupled to each other and laminated together. The semiconductor device **1** is, for example, a laminated image sensor, in which the first substrate **100** is provided with a pixel section **110** and the second substrate **200** is provided with a logic circuit **210** and a memory section **220**. In the present embodiment, the logic circuit **210** is provided on a surface (first surface, surface S**1**), of the second substrate, facing the first substrate **100**, and the memory section **220** is provided on a surface (second surface, surface S**2**), of the second substrate **200**, opposite to the surface facing the first substrate **100**.

Unit pixels are two-dimensionally arranged on the pixel section **110** of the first substrate **100**, in which there are disposed, for example, a backside illumination imaging element (imaging element **10**, see FIG. **2**), a transfer transistor that transfers charges obtained by photoelectric conversion of the imaging element **10** to an FD (floating diffusion) section, a reset transistor that resets an electric potential of the FD section, an amplifying transistor that outputs a signal corresponding to the electric potential of the FD section, and the like. The imaging element **10** corresponds to a specific example of an active element of the present disclosure.

As described above, the second substrate **200** is provided with the logic circuit **210** such as a control circuit that controls operation of the imaging element **10** on side of the surface S**1**, and a non-volatile memory element (storage element **40**) that configures the memory section **220** on side of the surface S**2**. It is to be noted that, on the surface S**1** side, in addition to the logic circuit, there may be disposed, for example, a circuit having an image processing function, an ADC (Analog digital converter) circuit that converts an analog signal outputted from the unit pixel provided on the pixel section into a digital signal and outputs the digital signal, and the like.

FIG. **2** illustrates an example of a specific cross-sectional configuration of the semiconductor device **1** illustrated in FIG. **1**. In the semiconductor device **1**, the first substrate **100** is provided with the imaging element **10**, as described above. The imaging element **10** has a configuration in which, for example, a planarization layer **14**, a color filter **15**, and a microlens **16** are laminated in this order on a semiconductor substrate **13** having a photodiode **13**A and a transistor **13**B embedded therein. The first substrate **100** includes a protective layer **17** on the microlens **16** of the imaging element **10**, and a glass substrate **18** is disposed on the protective layer **17**. Moreover, the first substrate **100** includes an electrically-conductive film **11** including Cu, for example, in the lowermost layer (a surface facing the second substrate **200**), and an insulating layer **12** is provided around the electrically-conductive film **11**.

The second substrate **200** includes a transistor **20** that configures the logic circuit **210** such as the control circuit on side of a surface **21**S**1** of a semiconductor substrate **21** (the surface S**1** side of the second substrate **200** in FIG. **1**), for example, that is on side of the surface facing the first substrate **100**. The transistor **20** is a transistor having a three-dimensional structure, for example, and is a Fin-FET transistor, for example.

FIG. **3** perspectively illustrates a configuration of the Fin-FET transistor **20**. The transistor **20** is configured by a fin **21**A, a gate insulating film **23**, and a gate electrode **24**. The fin **21**A includes Si, for example, and has a source region **21**S and a drain region **21**D. The fin **21**A is in a tabular form, and a plurality of fins **21**A stands on the semiconductor substrate **21** that includes, for example, Si. Specifically, the plurality of fins **21**A each extends in an X-axis direction, for example, and is arranged in parallel in a Z-axis direction. Provided on the semiconductor substrate **21** is an insulating layer **22** configured by SiO_2, for example. A portion of the fin **21**A is embedded in the insulating layer **22**. The gate insulating film **23** is provided to cover a side surface and an upper surface of the fin **21**A exposed from the insulating layer **22**, and is configured by, for example, HfSiO, HfSiON, TaO, TaON, or the like. The gate electrode **24** so extends in the Z direction that crosses the extending direction of the fin **21**A (X direction) as to straddle the fin **21**A. The fin **21**A is provided with a channel region **21**C at a portion crossing the gate electrode **24**, and the source region **21**S and the drain region **21**D are provided with the channel region **21**C interposed therebetween. It is to be noted that the cross-sectional structure of the transistor **20** illustrated in FIG. **2** illustrates a cross section taken along a line I-I in FIG. **3**.

The transistor **20** may be a Tri-Gate transistor, a nano-wire (Nano-Wire) transistor, an FD-SOI transistor, and a T-FET, other than the Fin-FET transistor described above. An inorganic semiconductor such as germanium (Ge) or a compound semiconductor such as a III-V semiconductor and a II-VI semiconductor, other than silicon (Si), may be used, as a semiconductor material, for the above-described transistors including the Fin-FET transistor. Specific examples of the II-VI semiconductor include InGaAs, InGaSb, SiGe, GaAsSb, InAs, InSb, InGanZnO (IGZO), MoS_2, WS_2, Boron Nitride, and Silicane Germanene. Another example thereof includes a graphene transistor using graphene. Moreover, the transistor **20** may be a transistor that employs a high dielectric constant film/metal gate (High-K/Metal Gate) technology. Alternatively, the transistor **20** may be a so-called Si planar transistor (see FIG. **8**).

A multilayer wiring forming section **31** is provided on the transistor **20**. The multilayer wiring forming section **31** includes, for example, an interlayer insulating film **32**, an interlayer insulating film **33**, and an interlayer insulating film **34** laminated in this order from side closer to the transistor **20**. A wiring line **31**A is provided for each of the interlayer insulating film **32**, the interlayer insulating film **33**, and the interlayer insulating film **34**. The wiring line **31**A is configured by a metal film M**1**, a metal film M**2**, and a metal film M**3** provided for the interlayer insulating films **32**, **33**, and **34**, respectively, and by a via V**1** and a via V**2** coupling them together. The via VI penetrates the interlayer insulating film **32** to couple the metal film M**1** and the metal film M**2** together. The via V**2** penetrates the interlayer insulating film **33** to couple the metal film M**2** and the metal film M**3** together. The metal film M**1**, the metal film M**2**, the metal film M**3**, the via V**1**, and the via V**2** include, for example, copper (Cu). The metal film M**3** is provided at the uppermost layer of the second substrate **200** (the surface facing the first substrate **100**). The first substrate **100** and the second substrate **200** are electrically coupled together by joining the electrically-conductive film **11** provided at the lowermost layer of the first substrate **100** and the metal film M**3**. It is to be noted that the configuration of the multilayer wiring forming section **31** illustrated in FIG. **2** is exemplary, and this is not limitative.

The storage element **40** configuring the memory section **220** is provided on side of a surface **21**S**2** of the semiconductor substrate **21** (the surface S**2** side of the second substrate **200** in FIG. **1**). The storage element **40** is, for example, a magnetic tunnel junction (Magnetic Tunnel Junction; MTJ) element. The storage element **40** has a configuration in which, for example, an electrically-conductive film **41**, a storage section **42**, and an electrically-conductive film **43** (which also serves as a bit line BL) are laminated in order via an insulating layer **35**. It is to be noted that the electrically-conductive film **41** is set as a lower electrode and the electrically-conductive film **43** is set as an upper electrode. The electrically-conductive film **41** is coupled to the source region **21**S or the drain region **21**D of the transistor **20** via a contact plug P**1**, for example. The contact plug P**1** has a truncated pyramid shape or a truncated cone shape, for example, with its occupied area increasing from the surface Si side toward the surface S**2** side (i.e., from a lower end toward an upper end) in this example. An insulating layer **36** is provided around the storage element **40**. The insulating layer **35** is configured by, for example, a High-K (high dielectric constant) film that allows for low-temperature formation, i.e., hafnium (Hf) oxide, aluminum oxide (Al_2O_3), ruthenium (Ru) oxide, tantalum (Ta) oxide, an oxide containing aluminum (Al), Ta, ruthenium (Ru), or Hf, and Si, a nitride containing Al, Ru, Ta, or Hf, and Si, an oxynitride containing Al, Ru, Ta. or Hf. and Si, or the like. The insulating layer **36** is configured by, for example, SiO_2 or a low-K (low dielectric constant) film.

The storage section **42** is preferably a spin-injection magnetization inverting storage element (STT-MTJ; Spin Transfer Torque-Magnetic Tunnel Junctions) that stores information by inverting an orientation of magnetization of a storage layer (storage layer **42**D, see FIG. **4**) described later by spin injection. The STT-MTJ is viewed as a promising non-volatile memory that replaces a volatile memory because the STT-MTJ allows for fast read/write.

The electrically-conductive film **41** and the electrically-conductive film **43** are each configured by, for example, a film of metal such as Cu, Ti, W, and Ru. The electrically-conductive film **41** and the electrically-conductive film **43** are each preferably configured by a metal film other than a constituent material of an underlayer **42**A or a cap layer **42**E described later, i.e., mainly include a Cu film, an Al film, or a W film. Moreover, the electrically-conductive film **41** and the electrically-conductive film **43** may also be configured as a metal film (single layer film) or a laminated film containing titanium (Ti), TiN (titanium nitride), Ta, TaN, (tantalum nitride), tungsten (W), Cu, Al, or the like.

FIG. **4** illustrates an example of a configuration of the storage section **42**. The storage section **42** has a configuration in which, for example, the underlayer **42**A, a magnetization pinned layer **42**B, an insulating layer **42**C, the storage layer **42**D, and the cap layer **42**E are laminated in this order from side closer to the electrically-conductive film **41**. Namely, the storage element **40** has a bottom pin structure including the magnetization pinned layer **42**B, the insulating layer **42**C, and the storage layer **42**D in this order upward from the bottom in a lamination direction. In the storage element **40**, information is stored by changing the orientation of magnetization M**42**D of the storage layer **42**D having a uniaxial anisotropy, and "0" or "1" of the information is defined by a relative angle (parallel or antiparallel) between the magnetization M**42**D of the storage layer **42**D and magnetization M**42**B of the magnetization pinned layer **42**B.

The underlayer **42**A and the cap layer **42**E are each configured by a metal film (single layer film) or a laminated film containing Ta, Ru, or the like.

The magnetization pinned layer **42**B is a reference layer serving as a reference of stored information (magnetization direction) in the storage layer **42**D, and configured by a ferromagnetic body having a magnetic moment with a direction of the magnetization M**42**B being pinned in a direction perpendicular to a film surface. The magnetization pinned layer **42**B is configured by, for example, cobalt (Co)-iron (Fe)-boron (B).

It is not desirable that the direction of the magnetization M**42**B of the magnetization pinned layer **42**B be changed by writing and reading; however, the direction may not necessarily be pinned in a specific direction. One reason for this is that it is sufficient to have less freedom of motion in the direction of the magnetization M**42**B of the magnetization pinned layer **42**B than the direction of the magnetization M**42**D of the storage layer **42**D. For example, it is sufficient for the magnetization pinned layer **42**B to have higher coercive force, a larger magnetic film thickness, or a larger magnetic damping constant than the storage layer **42**D. To pin the direction of the magnetization M**42**B, for example, an antiferromagnetic body such as PtMn and IrMn may be provided in contact with the magnetization pinned layer **42**B. Alternatively, the direction of the magnetization M**42**B may be indirectly pinned by magnetically coupling a magnetic body in contact with such an antiferromagnetic body to the magnetization pinned layer **42**B via a non-magnetic body such as Ru.

The insulating layer **42**C is an intermediate layer serving as a tunnel barrier layer (tunnel insulating layer), and is configured by, for example, Al_2O_3 or magnesium oxide (MgO). Among others, the insulating layer **42**C is preferably configured by MgO. This makes it possible to increase a magnetic resistance change rate (MR ratio), to improve efficiency of spin injection, and to reduce current density for inverting the orientation of the magnetization M**42**D of the storage layer **42**D.

The storage layer **42**D is configured by the ferromagnetic body having a magnetic moment that allows the direction of the magnetization M**42**D to freely change to the direction perpendicular to the film surface. The storage layer **42**D is configured by, for example, Co—Fe—B.

It is to be noted that, although the present embodiment has been described hereinabove with reference to the MTJ element as the storage element **40**, any other nonvolatile element may be used as well. Examples of the nonvolatile element other than the MTJ element include a variable resistance element such as a ReRAM and a FLASH.

Moreover, the surface Si of the second substrate **200** may be provided with a programmable circuit in addition to the control circuit. This makes it possible to change and automate operation of an imaging apparatus as needed.

(1-2. Method of Manufacturing Semiconductor Device)

The semiconductor device **1** of the present embodiment may be manufactured in the following manner, for example. FIGS. **5**A to **5**E illustrate an example of the method of manufacturing the semiconductor device **1** in a process order.

First, as illustrated in FIG. **5**A, the transistor **20** and the multilayer wiring forming section **31** that configure the logic circuit are formed in the second substrate **200**. Subsequently, the first substrate **100** provided with the imaging element **10** formed separately and the second substrate **200** are laminated by joining the electrically-conductive film **11** provided at the lowermost layer of the first substrate **100** and the metal film M**3** provided at the uppermost layer of the second substrate **200** to each other. Next, as illustrated in FIG. **5**C, the color filter **15**, the microlens **16**, and the protective layer **17** are formed on the planarization layer **14** of the imaging element **10** of the first substrate **100**, and thereafter the glass substrate **18** is adhered onto the protective layer **17**.

Subsequently, as illustrated in FIG. **5**D, the entire device is inversed using the glass substrate **18** as a supporting substrate, and the semiconductor substrate **21** of the second substrate **200** is polished to make the device thinner. Next, as illustrated in FIG. **5**E, the contact plug P**1** and the storage element **40** are formed. The contact plug P**1** couples the source region **21**S of the transistor **20** and the storage element **40** to each other via the insulating layer **35**, for example. This allows the semiconductor device **1** illustrated in FIG. **2** to be complete.

(1-3. Workings and Effects)

The MTJ element using the magnetic material is viewed as a promising non-volatile memory that replaces the volatile memory. However, as described above, the MTJ element has low resistance to heat, and this may possibly lead to degradation of the element characteristics due to the thermal budget during the wiring forming process.

It is possible to avoid the degradation of the element characteristics due to the thermal budget by forming the MTJ element after finishing the wiring process. However, with the semiconductor device including the active element such as the image sensor laminated on the logic circuit, the MR ratio of the MTF element may possibly be degraded by the thermal budget during the lamination process of the image sensor.

Meanwhile, in the semiconductor device **1** of the present embodiment, the imaging element **10** (pixel section **110**) is provided in the first substrate **100**, and the logic circuit **210** including the control circuit of the imaging element **10** and the storage element **40** (memory section **230**) are provided in the second substrate **200**. In particular, the transistor **20** configuring the logic circuit **210** is provided on the surface Si side of the second substrate **200**, and the storage element **40** is provided on the surface **S2** side of the second substrate **200**. This makes it possible to form the storage element **40** at a desired timing, i.e., specifically after forming the transistor **20** and the imaging element **10** and joining the first substrate **100** and the second substrate together. Thus, it is possible to reduce the thermal budget to the storage element **40**.

As described above, in the present embodiment, the imaging element **10** is provided in the first substrate **100**, the transistor **20** configuring the logic circuit **210** is provided on the surface **S1** side of the second substrate **200**, and the storage element **40** is provided on the surface **S2** side opposite to the surface **S2**. This makes it possible to form the storage element **40** after the process of forming a wiring line including the transistor **20** and a process of forming the imaging element **10**, which makes it possible to reduce the thermal budget to the storage element **40** and to prevent the degradation of the element characteristics.

It is to be noted that, although the present embodiment has been described with reference to the image sensor (imaging element **10**) as an example of the active element, this is not limitative; for example, various sensor functions such as a temperature sensor, a gravity sensor, and a position sensor may be included.

Next, second and third embodiments and Modification Examples 1 and 2 are described. It is to be noted that components corresponding to those in the semiconductor device **1** of the above-described first embodiment are denoted with the same reference numerals.

2. Second Embodiment

FIG. **6** illustrates a schematic configuration of a semiconductor device (semiconductor device **2**) according to a second embodiment of the present disclosure. The semiconductor device **2** is a laminated image sensor, and includes the first substrate **100**, the second substrate **200**, and a third substrate **300** electrically coupled to one another and laminated together. The semiconductor device **2** has a configuration in which the third substrate **300** is disposed between the first substrate **100** and the second substrate **200**. In the semiconductor device **2** of the present embodiment, similarly to the first embodiment, the imaging element **10** is provided in the first substrate **100**, and the storage element **40** is provided on the surface **S2** side of the second substrate **200**; among the circuits configuring the image sensor, circuits having mutually different supply voltages are provided separately on the surface S side of the second substrate **200** and on the third substrate **300**. Specifically, among the circuits provided in the semiconductor device **2**, a circuit having the lowest supply voltage is provided on the surface

Si of the second substrate **200**, and a circuit having the highest supply voltage is provided on the third substrate **300**.

Here, the circuit having the lowest supply voltage means a circuit that includes a transistor having the lowest drive voltage, and is the logic circuit **210**, for example. The transistor having the lowest drive voltage means a transistor manufactured using a state-of-the-art process, and is, for example, the Fin-FET transistor illustrated in FIG. **3**, the Tri-Gate transistor, the nano-wire (Nano-Wire) transistor, the FD-SOI transistor, and the T-FET, or the transistor that employs the high dielectric constant film/metal gate (High-K/Metal Gate). Moreover, a functional block that allows for high-speed signal processing may be provided on the surface S1 of the second substrate **200**.

The circuit having the highest supply voltage means a circuit that includes a transistor having the highest drive voltage; for example, an analog circuit such as an ADC **310**, an input/output (Input/Output (I/O)) circuit **320**, a circuit for controlling operation of the imaging element **10**, and the like are provided. Moreover, for example, in a case where the circuit configuring the memory section **220** includes the transistor having the highest drive voltage, a circuit part (Non-volatile memory (NVM) circuit **330**) including a transistor that is driven at the highest voltage may be provided on the third substrate **300**. Here, the transistor having the highest drive voltage means a transistor manufactured using the existing manufacturing process, and is, for example, the Si planar transistor.

FIG. **7** illustrates an example of a specific cross-sectional configuration of the semiconductor device **2** illustrated in FIG. **6**. The semiconductor device **2** includes the first substrate **100** described in the first embodiment. In the present embodiment, the second substrate **200** is provided with the logic circuit **210** including the Fin-FET transistor **20**, for example, and the third substrate **300** is provided with the ADC **310** including a transistor (transistor **60**) having an Si planar structure, the I/O circuit **320**, and the NVM **330**.

The third substrate **300** includes, for example, a multilayer wiring forming section **70** and a surface wiring forming section **75** laminated in this order on a surface **50**S2 of a semiconductor substrate **50**. The Si planar transistor **60** is provided near the surface **50**S2 of the semiconductor substrate **50**, and an electrically-conductive film **54** is provided on the surface Si side of the semiconductor substrate **50** with insulating layers **52** and **53** interposed therebetween. It is to be noted that, although FIG. **7** illustrates an example where three transistors **60** are provided, the number of the transistors **60** provided on the semiconductor substrate **50** is not particularly limited. The number may be one, or two or more. Moreover, a transistor other than the Si planar transistor may be provided.

The semiconductor substrate **50** is provided with an element separation film **51** formed by, for example, STI (Shallow Trench Isolation). The element separation film **51** is an insulating film including a silicon oxide film (SiO_2), for example, and one surface thereof is exposed to the surface **50**S2 of the semiconductor substrate **50**.

The semiconductor substrate **50** has a laminated structure of a first semiconductor layer **50**A (hereinbelow, referred to as a semiconductor layer **50**A) and a second semiconductor layer **50**B (hereinbelow, referred to as a semiconductor layer **50**B). The semiconductor layer **50**A has a configuration in which a channel region configuring a portion of the transistor **60** and a pair of diffusion layers **62** (described later) are formed on monocrystalline silicon, for example. The semiconductor layer **50**B includes the monocrystalline silicon, for example, and has a polarity different from that of the semiconductor layer **50**A. The semiconductor layer **50**B is provided to cover the semiconductor layer **50**A and the element separation film **51**.

A surface (a surface on side facing the first substrate **100**) of the semiconductor layer **50**B is covered with the insulating layer **52**. The semiconductor layer **50**B has an opening **50**K. The opening **50**K is filled with the insulating layer **52**. Furthermore, a contact plug P**2** extending to penetrate a coupling part of the insulating layer **52** and the element separation film **51**, for example, is provided at the opening **50**K part. The contact plug P**2** is formed by using a material mainly including, for example, low-resistance metal such as Cu, W, or Al. Moreover, it is preferable that a barrier metal layer including Ti or Ta alone or an alloy thereof be provided around the low-resistance metal. A periphery of the contact plug P**2** is covered with the element separation film **51** and the insulating layer **52**, and is electrically separated from the semiconductor substrate **50** (the semiconductor layer **50**A and the semiconductor layer **50**B).

The transistor **60** is the Si planar transistor, and includes, for example, a gate electrode **61** and the pair of diffusion layers **62** (**62**S and **62**D) serving as a source region and a drain region, as illustrated in FIG. **8**. Moreover, the transistor **60** provided on the semiconductor substrate **50** is covered with an interlayer insulating film **66** and is embedded in an interlayer insulating film **67**.

The gate electrode **61** is provided on the surface **50**S2 of the semiconductor substrate **50**. However, a gate insulating film **63** including a silicon oxide film or the like is provided between the gate electrode **61** and the semiconductor substrate **50**. It is to be noted that the thickness of the gate insulating film **63** is larger than that of the transistor having the three-dimensional structure such as the Fin-FET described above. Provided on a side surface of the gate electrode **61** is a side wall **64** including a laminate film of a silicon oxide film **64**A and a silicon nitride film **64**B, for example.

The pair of diffusion layers **62** include silicon in which impurities are diffused, for example, and configure the semiconductor layer **50**A. Specifically, the pair of diffusion layers **62** include a diffusion layer **62**S corresponding to the source region and a diffusion layer **62**D corresponding to the drain region, and are disposed with the channel region interposed therebetween. The channel region is facing the gate electrode **61** in the semiconductor layer **50**A. Portions of the diffusion layers **62** (**62**S, **62**D) are provided with silicide regions **65** (**65**S, **65**D) including, for example, metal silicide such as nickel silicide (NiSi) or cobalt silicide (CoSi). The silicide regions **65** reduces contact resistance between each of connections **68**A to **68**C described later and the diffusion layers **62**. The silicide region **65** has one surface thereof that is exposed to the surface **50**S2 of the semiconductor substrate **50**, whereas an opposite surface thereof is covered with the semiconductor layer **50**B. Moreover, the thickness of each of the diffusion layers **62** and the silicide region **55** is smaller than that of the element separation film **51**.

The interlayer insulating film **67** is provided with the connections **68**A to **68**C penetrating the interlayer insulating film **66** along with the interlayer insulating film **67**. The silicide region **65**D of the diffusion layer **62**D serving as the drain region and the silicide region **65**S of the diffusion layer **62**S serving as the source region are coupled to a metal film M**1'** of a wiring line **70**A described later via the connection **68**B and the connection **68**C, respectively. The contact plug P**2** penetrates the interlayer insulating films **66** and **67**, and contacts, at a lower end thereof, the metal film M**1'** configuring a selection line SL, for example. Thus, the contact plug **P2** extends to penetrate all of the insulating layer **52**, the element separation film **51**, the interlayer insulating film **66**, and the interlayer insulating film **67**.

The multilayer wiring forming section **70** includes, for example, an interlayer insulating film **71**, an interlayer insulating film **72**, an interlayer insulating film **73**, and an interlayer insulating film **74** laminated in order from side closer to the transistor **60**. The wiring line **70**A is provided for each of the interlayer insulating film **71**, the interlayer insulating film **72**, the interlayer insulating film **73**, and the interlayer insulating film **7**. The wiring line **70**A is configured by the metal film M**1'**, a metal film M**2'**, a metal film M**3'**, a metal film M**4'**, and a metal film M**5**, as well as vias V**1'**, V**2'**, V**3'**, V**4'**, and V**5'** that couple them together. Here, the metal film M**1'**, the metal film M**2'**, the metal film M**3'**, and the metal film M**4'** and the metal film M**5** are respectively embedded in the interlayer insulating film **71**, the interlayer insulating film **72**, the interlayer insulating film **73**, and the interlayer insulating film **74**. Moreover, the metal film M**1'** and the metal film M**2'** are coupled by the via V**1'** that penetrates the interlayer insulating film **71**. Similarly, the metal film M**2'** and the metal film M**3'** are coupled by the via V**2'** penetrating the interlayer insulating film **72**. The metal film **3'** and the metal film M**4'** are coupled by the via V**3'** penetrating the interlayer insulating film **73**. The metal film **4'** and the metal film M**5'** are coupled by the via V**4'** penetrating the interlayer insulating film **74**. As described above, the wiring line **70**A is coupled to the diffusion layers **62** serving as the drain region and the source region of the transistor **60**, respectively, via the connection **68**B and the connection **68**C in contact with the metal film M**1'**. It is to be noted that the configuration of the multilayer wiring forming section **70** illustrated in FIG. **7** is exemplary, and this is not limitative.

Provided on the multilayer wiring forming section **70** is the surface wiring forming section **75** to be joined to the second substrate **200**. In the surface wiring forming section **75**, a metal film **77** including Cu, for example, is embedded in a surface of an insulating layer **76**, and the surface of the metal film **77** is exposed to the insulating layer **76**. The second substrate **200** and the third substrate **300** are electrically coupled by joining the metal film **77** and the metal film M**3** of the second substrate **200** together. The metal film **77** is coupled to the metal film M**5'** of the multilayer wiring forming section **70** via the via V**5'** penetrating the insulating layer **76**.

Provided on a surface **50**S**1** of the semiconductor substrate **50** is the insulating layer **52**. The insulating layer **52** is configured by a High-K film allowing for low-temperature formation, for example. The insulating layer **53** is laminated on the insulating layer **52**. The insulating layer **53** is configured by a film of a material (Low-K) having a specific dielectric constant lower than that of SiO_2, for example. Examples of the High-K film allowing for low-temperature formation include Hf oxide, Al_2O_3, Ru oxide, Ta oxide, an oxide containing Al, Ru. Ta. or Hf, and Si, a nitride containing Al, Ru, Ta, or Hf, and Si, an oxynitride containing Al., Ru, Ta, or Hf. and Si, and the like. The electrically-conductive film **54** is provided on first substrate **100** side of the insulating layer **53**, and a surface of the electrically-conductive film **54** is exposed. The electrically-conductive film **54** is in contact with an upper end of the contact plug P**2**, and is joined to the electrically-conductive film **11** on the opposite surface. The electrically-conductive film **11** is provided at the lowermost layer of the first substrate **100**. This allows the first substrate **100** and the third substrate **300** to be electrically coupled to each other.

As described above, in the semiconductor device **2** according to the present embodiment, circuits having different supply voltages, among a plurality of circuits configuring the imaging apparatus, are separately provided on separate substrates (the second substrate **200** and the third substrate **300**). Specifically, a circuit including a transistor having the lowest drive voltage, as with the logic circuit **210**, is provided on the surface SI of the second substrate **20**X), and a circuit including a transistor having the highest drive voltage is provided on the third substrate **300**.

This achieves an effect of making it possible to reduce the size of the semiconductor device, in addition to the effects of the above-described first embodiment. Moreover, providing the circuits having different supply voltages on separate substrates allows the transistor (the transistor **20** in this example) using the state-of-the-art process, for example, as described in the above-described first embodiment, for example, and the transistor (the transistor **60** in this example) using the existing manufacturing process to be provided on different substrates. This simplifies the manufacturing process, thus achieving effects of making it possible to reduce manufacturing cost and to improve manufacturing yield.

3. Modification Example 1

FIG. **9** illustrates an example of a specific cross-sectional configuration of a semiconductor device (semiconductor device **3**) according to a modification example (Modification Example 1) of the second embodiment of the present disclosure. The present modification example is different from the above-described second embodiment in that a leading electrode **80** is provided on the surface S**2** side of the second substrate **200**.

The leading electrode **80** is configured by an electrically-conductive film **82** and a bump **84**. The electrically-conductive film **82** is provided on a back surface (surface **21**S**2**) of the semiconductor substrate **21** with insulating layers **35** and **36** and an insulating layer **81** interposed therebetween. The insulating layer **81** is configured by, for example, an SiO_2 film. An insulating layer **83** configured by the SiO_2 film, for example is provided around the electrically-conductive film **82**. The electrically-conductive film **82** has a configuration in which, for example, an electrically-conductive film **82**A including Cu and an electrically-conductive film **82**B including Al are laminated in this order. The leading electrode **80** is electrically coupled to a wiring line **24**A via a contact plug P**3**. The contact plug P**3** penetrates the semiconductor substrate **21** and the insulating layers **22**, **35**, **36**, and **81**, for example. The wiring line **24**A is separated from the gate electrode **24** by an insulating film **85**, for example. It is to be noted that a periphery of the contact plug P**3** is preferably covered with the insulating film, as illustrated in FIG. **9**.

This makes it possible to configure an electrode outlet port anywhere even when the storage element **40** is provided on the surface S**2** side of the second substrate **200**.

It is to be noted that the leading electrode **80** is desirably formed at a temperature for forming the storage element **40** or lower because the leading electrode **80** is formed after forming the storage element **40**. Moreover, the leading electrode **80** is allowed to be formed by exposing the metal film to be the electrode not only on side of the surface **21**S**1** of the semiconductor substrate **21** but also on a side surface of the second substrate **200**, for example.

4. Third Embodiment

FIG. **10** illustrates a schematic configuration of a semiconductor device (semiconductor device **4**) according to the third embodiment of the present disclosure. The semiconductor device **4** is mounted with, as another example of the active element, a platform for communication applied to various frequency bands ranging from a near field to a far field, for example. A first substrate **400** includes an analog circuit **420** configuring the communication platform on side of a surface (surface S**3**), of the first substrate **400**, facing the third substrate **300**, for example. The second substrate **200** and the third substrate **300** each have a configuration similar to that of the above-described second embodiment.

As illustrated in FIG. **10**, the first substrate **400** includes the analog circuit **420** configuring the communication platform on the surface S**3** side facing the third substrate **300**. Specific examples of the analog circuit **420** include, for example, an RF front-end section including a transmit-receive switch and a power amplifier, and an RF-IC section including a low-noise amplifier and a transmit-receive mixer.

Although the first substrate **400** generally uses a silicon (Si) substrate as a core substrate, as described in the above-described first embodiment, there are cases in which a compound semiconductor substrate is partially used. For example, there are cases in which, for example, the above-described RF front-end section and RF-IC section are provided at a gallium nitride (GaN) substrate.

FIG. **11** illustrates an example of a specific cross-sectional configuration of the semiconductor device **4** illustrated in FIG. **10**. The present embodiment is described with reference to a case of using a GaN substrate **91** as the semiconductor substrate in the first substrate **400**.

The first substrate **400** is provided with a transistor **90** on a surface **91**S**3** of the GaN substrate **91**, for example. The transistor **90** is a high electron mobility transistor (High Electron Mobility Transistor; HEMT), for example. The HEMT is a transistor that controls two-dimensional electron gas (channel region **90**C) formed on a heterojunction interface including heterogenous semiconductor by means of a field effect. On the GaN substrate **91**, for example, an AlGaN layer **93** (or AlInN layer) is provided, which forms an AlGaN/GaN heterostructure. On the AlGaN layer **93**, a gate electrode **96** is provided with a gate insulating film **94** interposed therebetween. Moreover, on the AlGaN layer **93**, a source electrode **96**S and a drain electrode **96**D are provided with the gate electrode **96** interposed therebetween. The AlGaN layers **93** in contact with the source electrode **96**S and the drain electrode **96**D are each provided with an n-type region **93**N. An element separation film **95** is provided adjacent to the transistor **90**. An interlayer insulating film **97** is provided around the gate electrode **96**, the source electrode **96**S, and the drain electrode **96**D. Provided on the interlaver insulating film **97** is a multilayer wiring forming section including a metal film M**1**" and a metal film M**2**" laminated in this order from side closer to the transistor **90**. The metal film M**1**" and the metal film M**2**" are embedded in an interlayer insulating layer **98**, and the metal film M**1**" and the metal film M**2**" are coupled by a via V**1**" penetrating the interlayer insulating layer **98**. The first substrate **400** and the third substrate **300** are electrically coupled by joining the metal film M**2**" and the electrically-conductive film **54** together. Provided on the other surface (surface **91**S**4**) of the GaN substrate **91** is an Si substrate **92** as a base substrate.

In this manner, in the present embodiment, the first substrate **400** including a platform for communication is laminated on the second substrate **200** provided with the storage element **40**. Moreover, the storage element **40** is provided on the surface S**2** side of the second substrate **200**. This allows the storage element **40** to be formed after the wiring forming process of the second substrate **200** and the process of forming the platform for communication, thus making it possible to reduce the thermal budget to the storage element **40** and to prevent degradation of the element characteristics.

That is, regardless of the type of the active element, the circuit including the storage element **40** along with the transistor **20** and the active element (e.g., the imaging element **10** and the platform for communication) are provided on separate substrates (the first substrate **100** (, **400**) and the second substrate **200**). Furthermore, the circuit including the transistor **20** and the storage element **40** are provided on different surfaces (the surface Si and the surface S**2**) of the substrate (second substrate **200**). This makes it possible to provide the semiconductor device with reduced element characteristics of the storage element **40**.

It is to be noted that, although the present embodiment has been described with reference to an example in which three substrates (the first substrate **400**, the second substrate **200**, and the third substrate **300**) are laminated similarly to the second embodiment, the embodiment is also applicable to the semiconductor device configured by two substrates (the first substrate **400** and the second substrate **200**) as in the above-described first embodiment.

5. Modification Example 2

FIG. **12** illustrates a schematic configuration of a semiconductor device (semiconductor device **5**) according to a modification example of the third embodiment of the present disclosure. FIG. **13** illustrates an example of a specific cross-sectional configuration of the semiconductor device **5** illustrated in FIG. **12**. In the present modification example, for example, an antenna **920** is provided on side of a surface S**4** opposite to the surface S**3** of the first substrate **400**. Moreover, a shield structure (shield layer **910**) is provided between the transistor **90** provided on the surface S**3** side of the first substrate **400** and the antenna **920** provided on the surface S**4** side.

In the present modification example, the shield layer **910** is provided on the Si substrate **92** that is a base substrate of the surface **91**S**4** of the GaN substrate **91**, with an insulating layer **99**A interposed therebetween. The antenna **920** is disposed on the shield layer **910** with the insulating layer **99**B interposed therebetween. As a material of the shield layer **910**, it is preferable to use, for example, a magnetic material having a very small magnetic anisotropy and high initial permeability examples thereof include a permalloy material. An insulating layer **99**C is provided around the antenna **920**.

The antenna **920** is electrically coupled to the transmit-receive switch provided, for example, at the RF front-end section by means of the contact plug penetrating the GaN substrate **91**, for example, though not illustrated in FIG. **13**. The RF front-end section is provided on the surface S**3** side of the first substrate **400**, for example. The type of the antenna **920** is not particularly limited: examples thereof include a linear antenna such as a monopole antenna or a dipole antenna, and a planar antenna such as a microstrip antenna including a Low-K film interposed between metal films.

As described above, in the present modification example, the antenna **920** is provided on the surface S**4** of the first substrate **400**, thus making it possible to dispose, for example, the RF front-end section configuring the platform for communication provided on the surface S**3** and the antenna **920** at the shortest distance and couple them together. This makes it possible to perform a desired signal processing without attenuating signal intensity.

Moreover, providing the antenna **920** on a surface different from surfaces where various circuits such as the RF front-end section are provided increases design freedom, thus allowing for formation of the antenna **920** using suitable film thickness, size, or material. Accordingly, it is possible to improve element characteristics of the antenna **920**.

It is to be noted that, in addition to the antenna **920**, a capacitor, a coil, a resistor, or the like may be provided on the surface S**4** side of the first substrate **400**, as illustrated in FIG. **12**.

Although the present disclosure has been described hereinabove with reference to the first to third embodiments and Modification Examples 1 and 2, the present disclosure is not limited to the above-described embodiments, etc., and may be modified in a variety of ways. For example, although the above-described embodiments, etc. have been described with reference to specific configurations of the transistors **20** and **60**, the storage element **40**, and the like, there is no need to provide all of the components, and other components may be further provided.

Moreover, although the above-described embodiments, etc. have been described with reference to the semiconductor device including two or three substrates laminated together, four or more substrates may be laminated.

It is to be noted that the effects described in the present specification are merely exemplary and non-limiting, and other effects may be included.

Moreover, the semiconductor device of the present disclosure and the method of manufacturing the semiconductor device may have the following configurations.

(1)

A semiconductor device including:

a first substrate provided with an active element: and

a second substrate laminated with the first substrate and electrically coupled to the first substrate,

the second substrate being provided with a first transistor configuring a logic circuit on a first surface and with a non-volatile memory element on a second surface opposite to the first surface.

(2)

The semiconductor device according to (1), in which

the first transistor is provided on a surface, of the second substrate, facing the first substrate, and

the non-volatile memory element is provided on side opposite to the surface facing the first substrate.

(3)

The semiconductor device according to (1) or (2), further including a third substrate provided between the first substrate and the second substrate, the third substrate being provided with a second transistor that is driven at a drive voltage higher than a drive voltage of the first transistor.

(4)

The semiconductor device according to (3), in which the third substrate is provided with an analog circuit including the second transistor.

(5)

The semiconductor device according to any one of (1) to (4), in which the second substrate is provided with a leading electrode on the second surface.

(6)

The semiconductor device according to any one of (1) to (5), in which the non-volatile memory element includes a magnetic tunnel junction element.

(7)

The semiconductor device according to any one of (1) to (6), in which the active element includes an imaging element.

(8)

The semiconductor device according to any one of (1) to (7), in which the active element includes a circuit having a communication function.

(9)

The semiconductor device according to (8), in which

the first substrate includes the circuit having the communication function on a third surface facing the second substrate, and

the first substrate is provided with an antenna on a fourth surface opposite to the third surface.

(10)

The semiconductor device according to (9), further including a shield structure provided between the circuit having the communication function and the antenna.

(11)

The semiconductor device according to any one of (1) to (10), in which

the first substrate includes a core substrate, and

the core substrate includes a compound semiconductor substrate.

(12)

A method of manufacturing a semiconductor device, the method including:

forming an active element on a first substrate;

forming a transistor configuring a logic circuit on a first surface of a second substrate;

electrically coupling the first substrate and the second substrate together; and

forming a non-volatile memory element on a second surface, of the second substrate, opposite to the first surface.

(13)

The method of manufacturing the semiconductor device according to (12), the method further including:

joining the first substrate including the active element and the second substrate provided with the transistor to each other, with the first surface of the second substrate as a facing surface, and

thereafter forming the non-volatile memory element on the second surface of the second substrate.

(14)

The method of manufacturing the semiconductor device according to (13), the method further including forming a leading electrode on the second surface of the second substrate with an insulating layer interposed therebetween after the forming of the non-volatile memory element on the second surface.

(15)

The method of manufacturing the semiconductor device according to (14), in which the forming of the leading electrode includes forming the leading electrode at a temperature for forming the non-volatile memory element or lower.

This application claims the benefit of Japanese Priority Patent Application JP2017-020626 filed with the Japanese Patent Office on Feb. 7, 2017, the entire contents of which are incorporated herein by reference.

It should be understood by those skilled in the art that various modifications, combinations, sub-combinations, and alterations may occur depending on design requirements and other factors insofar as they are within the scope of the appended claims or the equivalents thereof.

What is claimed is:

1. A semiconductor device, comprising:
a first substrate including an active element;
a second substrate including:
a first surface;
a second surface;
a first transistor including:
a fin and
a gate insulating film provided to cover side surfaces and an upper surface of the fin,
wherein the first transistor configures a logic circuit provided on the first surface that controls operations of the active element; and
a non-volatile memory provided on the second surface opposite to the first surface,
wherein the second substrate is laminated with the first substrate and electrically coupled to the first substrate; and
a contact plug,
wherein the first transistor and the non-volatile memory are coupled together via the contact plug,
wherein the contact plug has a truncated pyramid shape with an occupied area of the truncated pyramid shape increasing from the first surface towards the second surface,
wherein the contact plug is provided between the first surface and the second surface of the second substrate without overlapping with the gate insulating film of the first transistor in a plan view along a length of the contact plug, and
wherein the contact plug electrically connects the first transistor and the non-volatile memory together.

2. The semiconductor device according to claim **1**, wherein
the first transistor is provided on a surface of the second substrate facing the first substrate, and
the non-volatile memory element is provided on a side opposite to the surface facing the first substrate.

3. The semiconductor device according to claim **1**, further comprising a third substrate provided between the first substrate and the second substrate, the third substrate being provided with a second transistor that is driven at a drive voltage higher than a drive voltage of the first transistor.

4. The semiconductor device according to claim **3**, wherein the third substrate is provided with an analog circuit including the second transistor.

5. The semiconductor device according to claim **1**, wherein the second substrate is provided with a leading electrode on the second surface.

6. The semiconductor device according to claim **1**, wherein the non-volatile memory element comprises a magnetic tunnel junction element.

7. The semiconductor device according to claim **1**, wherein the active element comprises an imaging element.

8. The semiconductor device according to claim **1**, wherein the active element comprises a circuit having a communication function.

9. The semiconductor device according to claim **8**, wherein
the first substrate includes the circuit having the communication function on a third surface facing the second substrate, and
the first substrate is provided with an antenna on a fourth surface opposite to the third surface.

10. The semiconductor device according to claim **9**, further comprising a shield structure provided between the circuit having the communication function and the antenna.

11. The semiconductor device according to claim **1**, wherein
the first substrate includes a core substrate, and
the core substrate comprises a compound semiconductor substrate.

12. A method of manufacturing a semiconductor device, the method comprising:
providing a first substrate;
forming an active element on the first substrate;
providing a second substrate including a first surface and a second surface;
forming a transistor including a fin and a gate insulating film provided to cover side surfaces and an upper surface of the fin,
wherein the first transistor configures a logic circuit on the first surface of the second substrate that controls operations of the active element;
electrically coupling the first substrate and the second substrate together;
providing a contact plug; and
forming a non-volatile memory element on the second surface of the second substrate opposite to the first surface,
wherein the transistor and the non-volatile memory are coupled together via the contact plug,
wherein the contact plug has a truncated pyramid shape with an occupied area of the truncated pyramid shape increasing from the first surface towards the second surface,
wherein the contact plug is provided between the first surface and the second surface of the second substrate without overlapping with the gate insulating film of the first transistor in a plan view along a length of the contact plug, and
wherein the contact plug electrically connects the transistor and the non-volatile memory together.

13. The method of manufacturing the semiconductor device according to claim **12**, the method further comprising:
joining the first substrate including the active element and the second substrate provided with the transistor to each other, with the first surface of the second substrate as a facing surface, and
thereafter forming the non-volatile memory element on the second surface of the second substrate.

14. The method of manufacturing the semiconductor device according to claim **13**, the method further comprising forming a leading electrode on the second surface of the second substrate with an insulating layer interposed therebetween after the forming of the non-volatile memory element on the second surface.

15. The method of manufacturing the semiconductor device according to claim **14**, wherein the forming of the leading electrode includes forming the leading electrode at a temperature for forming the non-volatile memory element or lower.

16. The semiconductor device according to claim **1**, wherein the second substrate further comprises a multilayer wiring section provided above the first transistor for electrically coupling the first substrate to the second substrate.

17. The semiconductor device according to claim **1**, wherein the first transistor comprises a fin field-effect transistor (FET).

18. The method of manufacturing the semiconductor device according to claim **12**, wherein the forming of the non-volatile memory element includes forming the non-volatile memory element using a magnetic tunnel junction element.

19. The method of manufacturing the semiconductor device according to claim **12**, the method further comprising forming a multilayer wiring section provided above the transistor for electrically coupling the first substrate to the second substrate.

20. The method of manufacturing the semiconductor device according to claim **12**, wherein the forming of the transistor includes forming the transistor using a fin field-effect transistor (FET).

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